

Evolutionary traits in Chikungunya virus untranslated regions

Michael T. Wolfinger

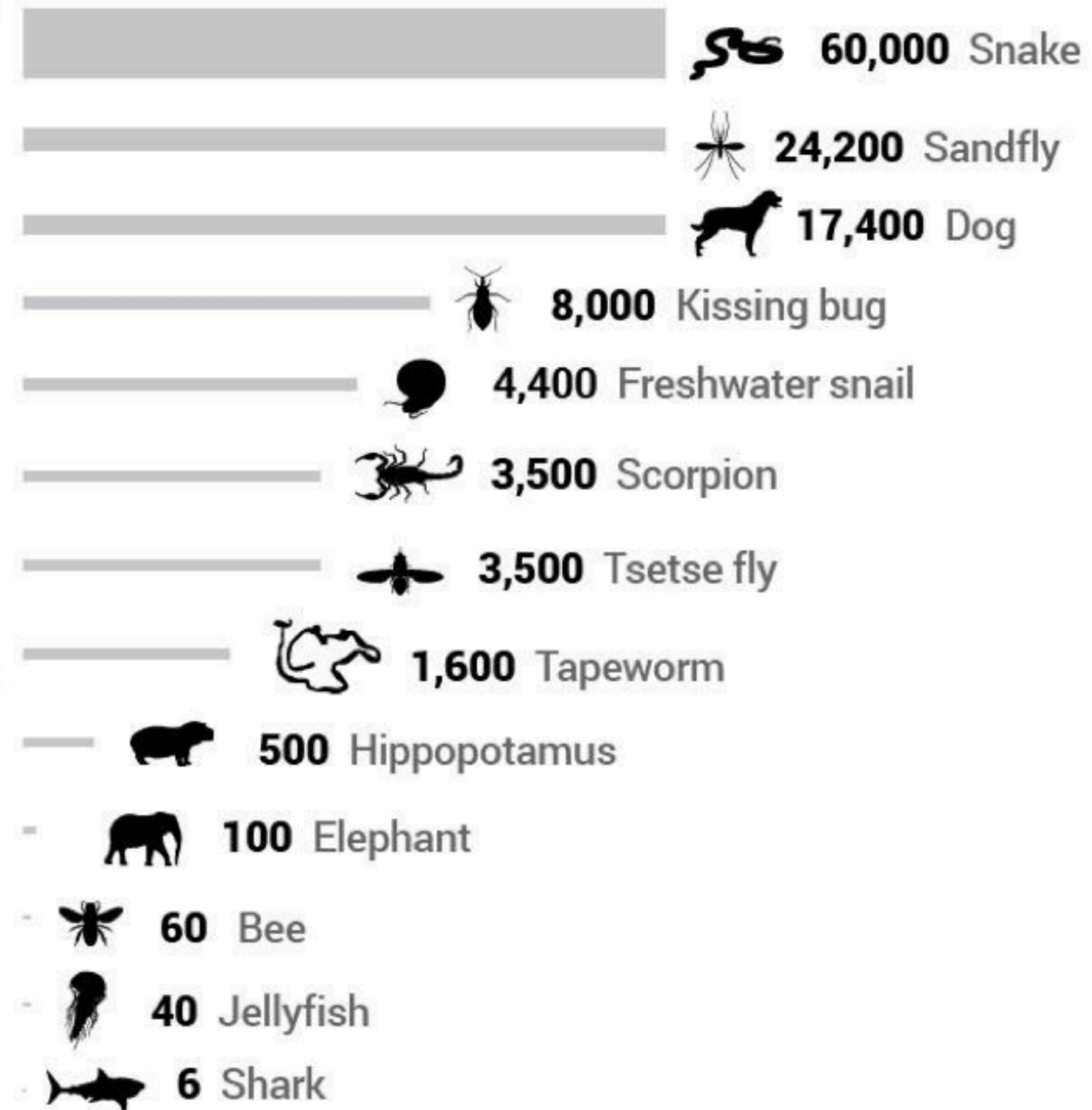
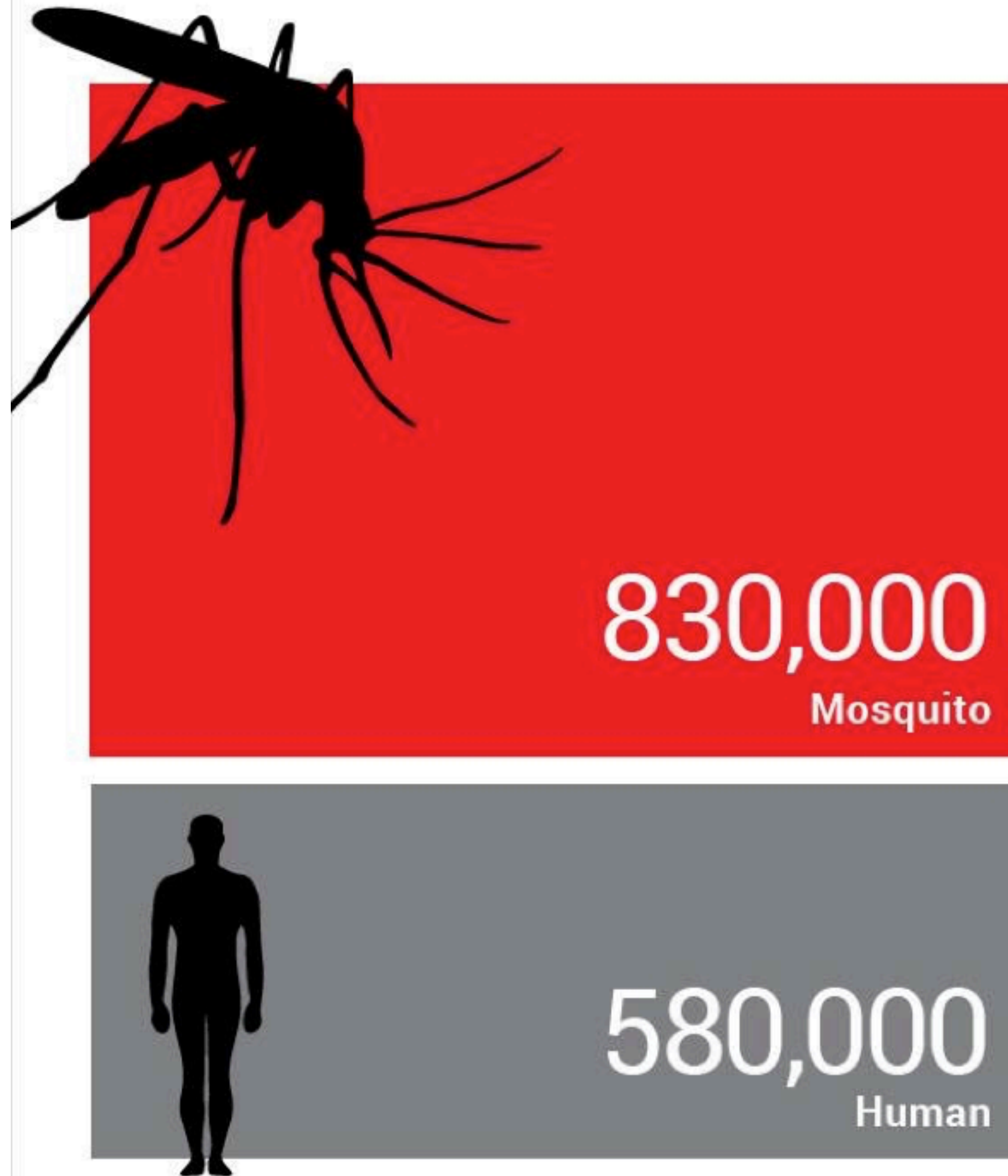
Research Group Bioinformatics and Computational Biology
& Department of Theoretical Chemistry
University of Vienna

4th VDS Retreat, Mondsee
26 June 2019

The World's Deadliest Animals

Number of people killed by animals, 2015

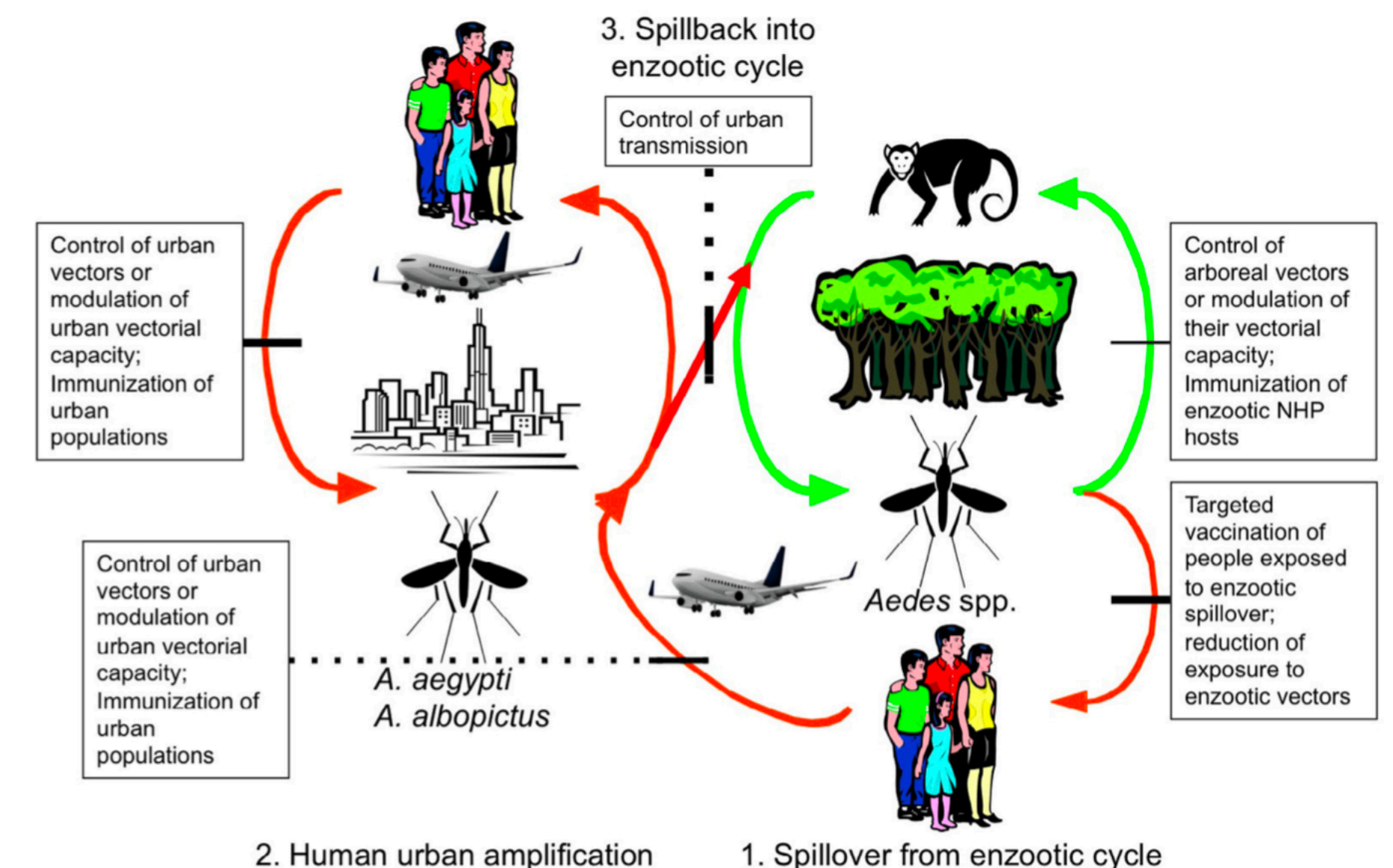
gates
notes



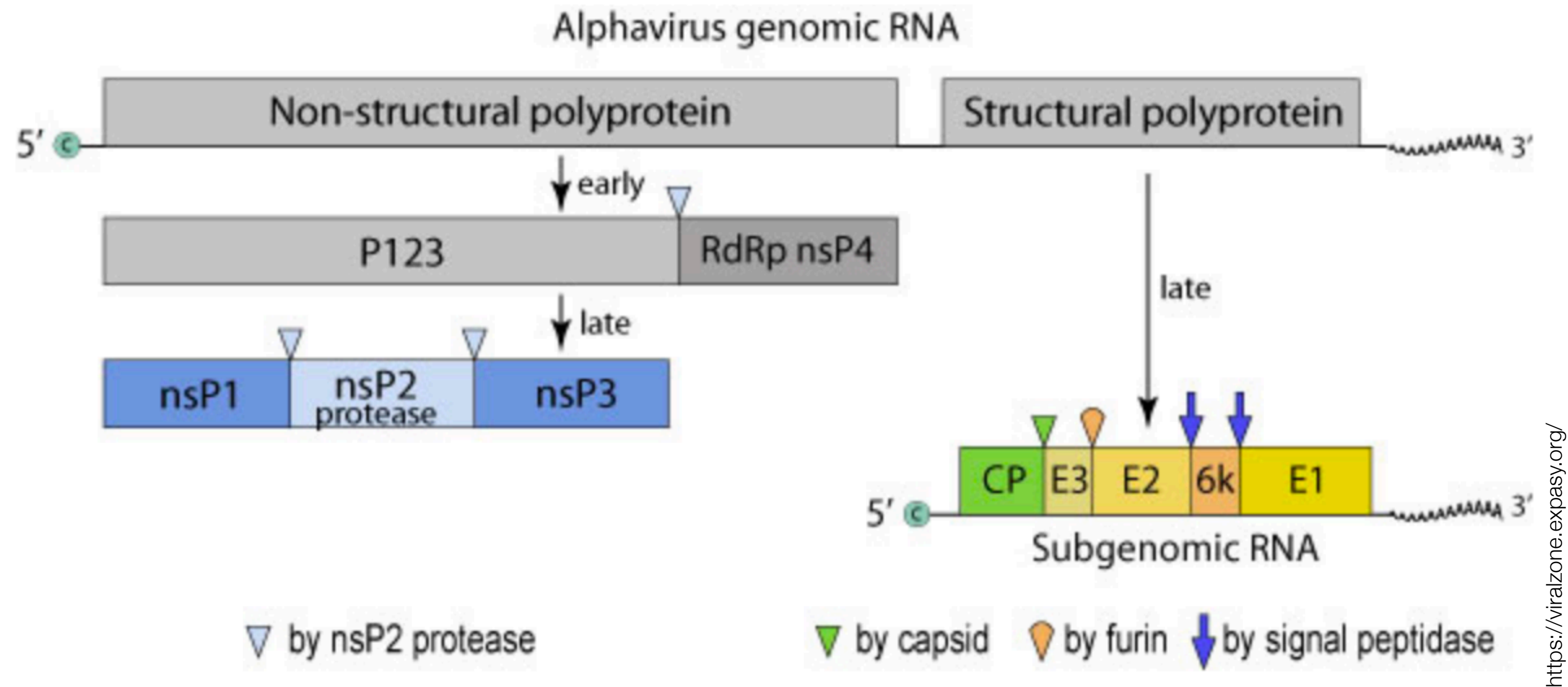
Sources: IHME, WHO, CrocBITE, FAO, Norwegian Institute for Nature Research, International Shark Attack File, National Geographic, PBS, National Science Foundation, CDC, WWF, Wilderness & Environmental Medicine, Nature, French Institute of Research for Development. All calculations have wide error margins.

Chikungunya Virus (CHIKV)

- Family *Togaviridae* / genus *Alphavirus*; mosquito-borne (*Aedes* spp.)
- Single-stranded (+) sense RNA virus
- “Chikungunya” (Makonde language): “Disease that bend up the joints”
- Chikungunya fever: febrile illness, arthralgia, rash, rarely causes hemorrhagic complications
- Enzootic in tropical and subtropical regions of Africa
- First outbreak described 1952 in Tanzania
- No vaccine available



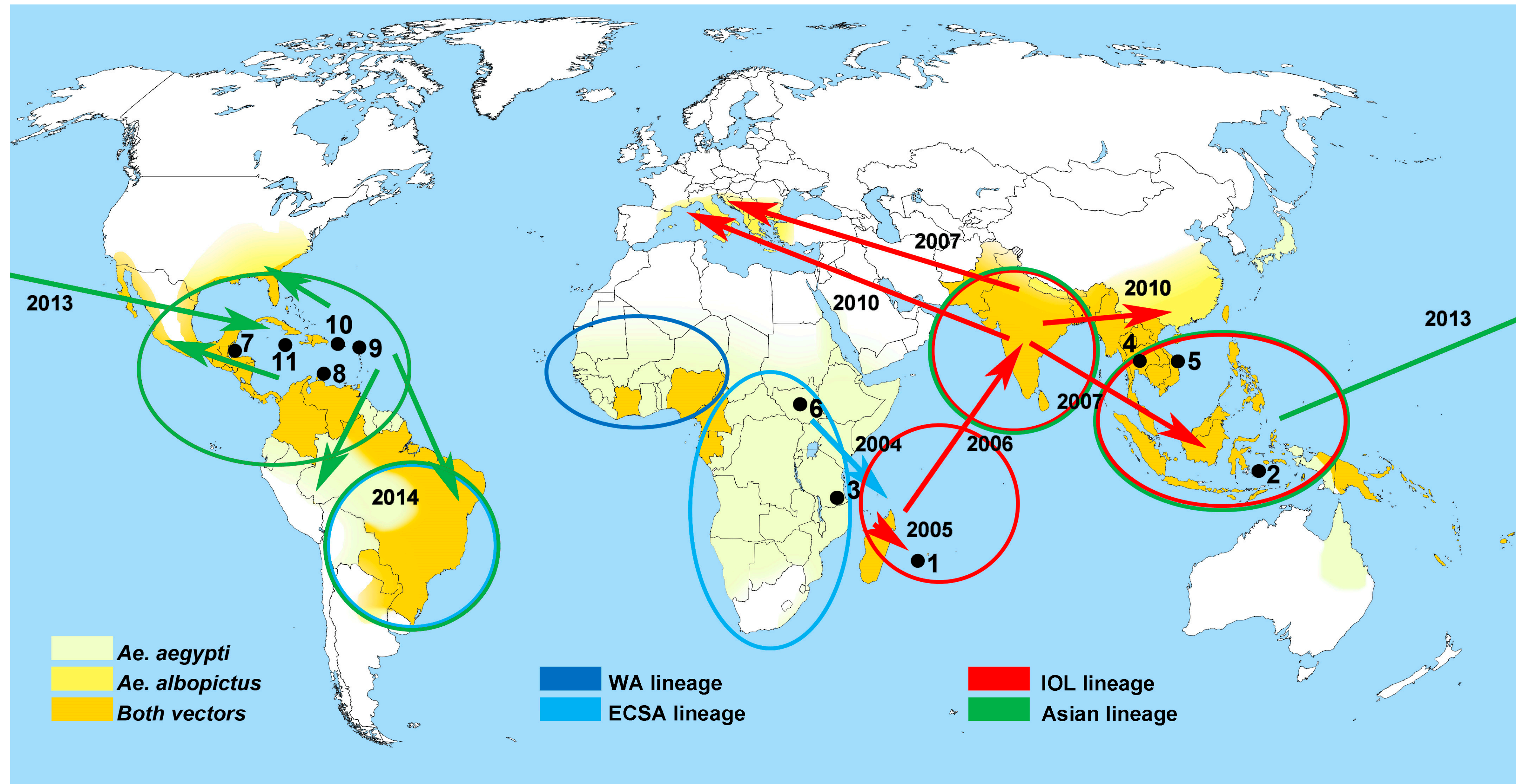
Alphavirus Genome Organization



- Non-segmented, single-stranded, (+)-sense RNA genomes of 11-12kB length
- Capped and polyadenylated
- AV genomes appear to host cells as mRNA for immediate translation upon entry into the cytoplasm

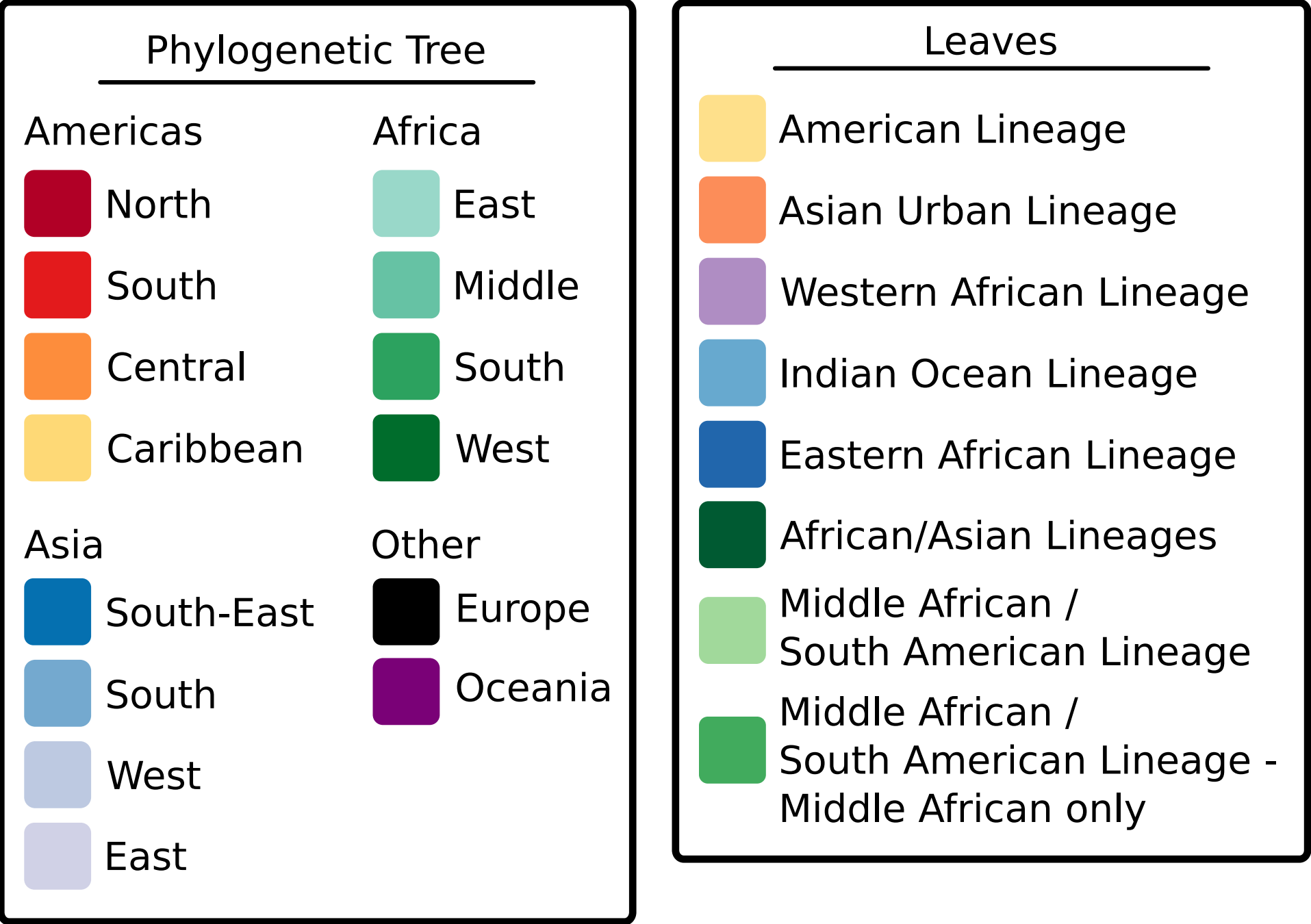
Part I: Phylogeography

CHIKV Epidemic Spread

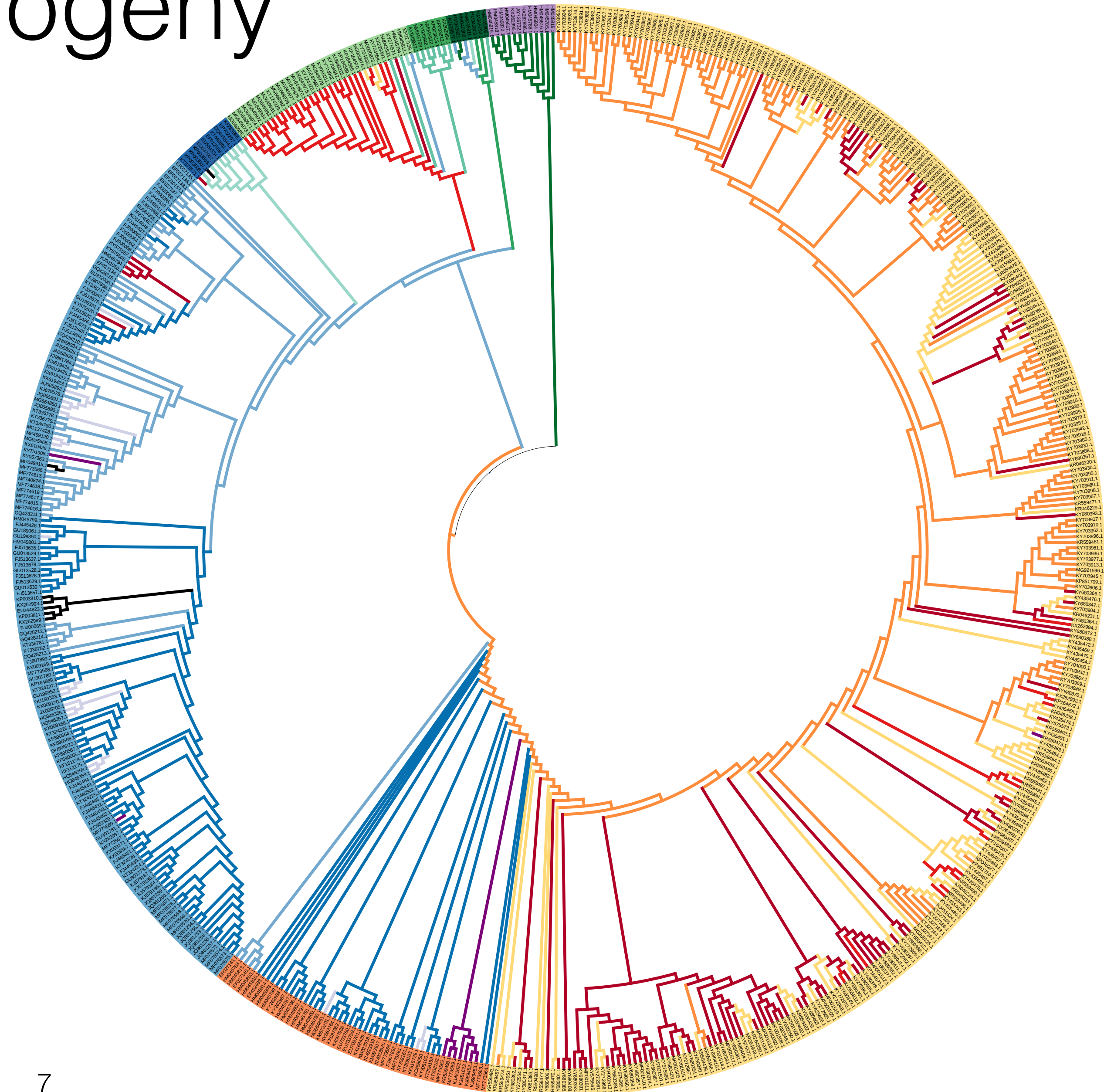


Frickmann et al., Viruses (2019)

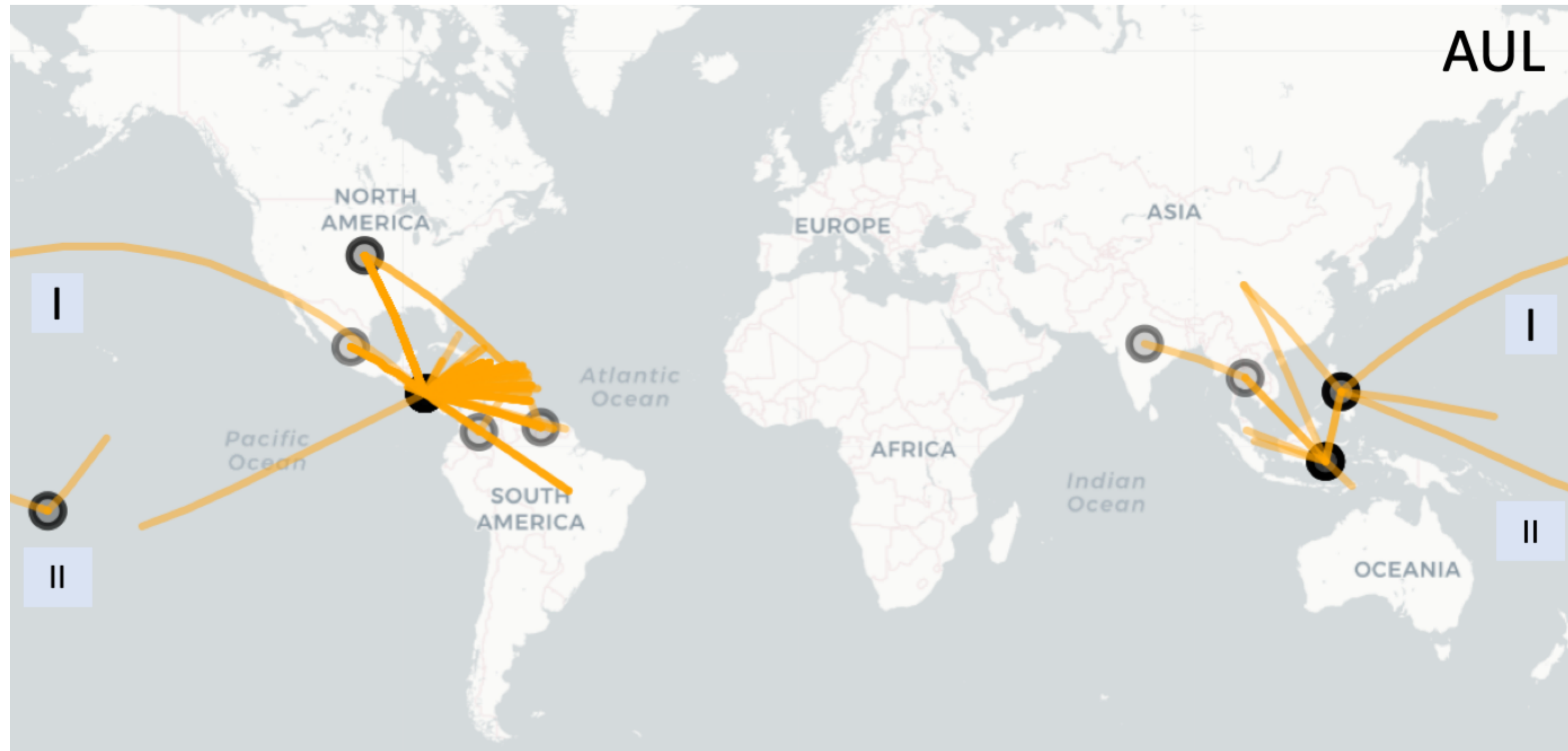
Updated CHIKV Phylogeny



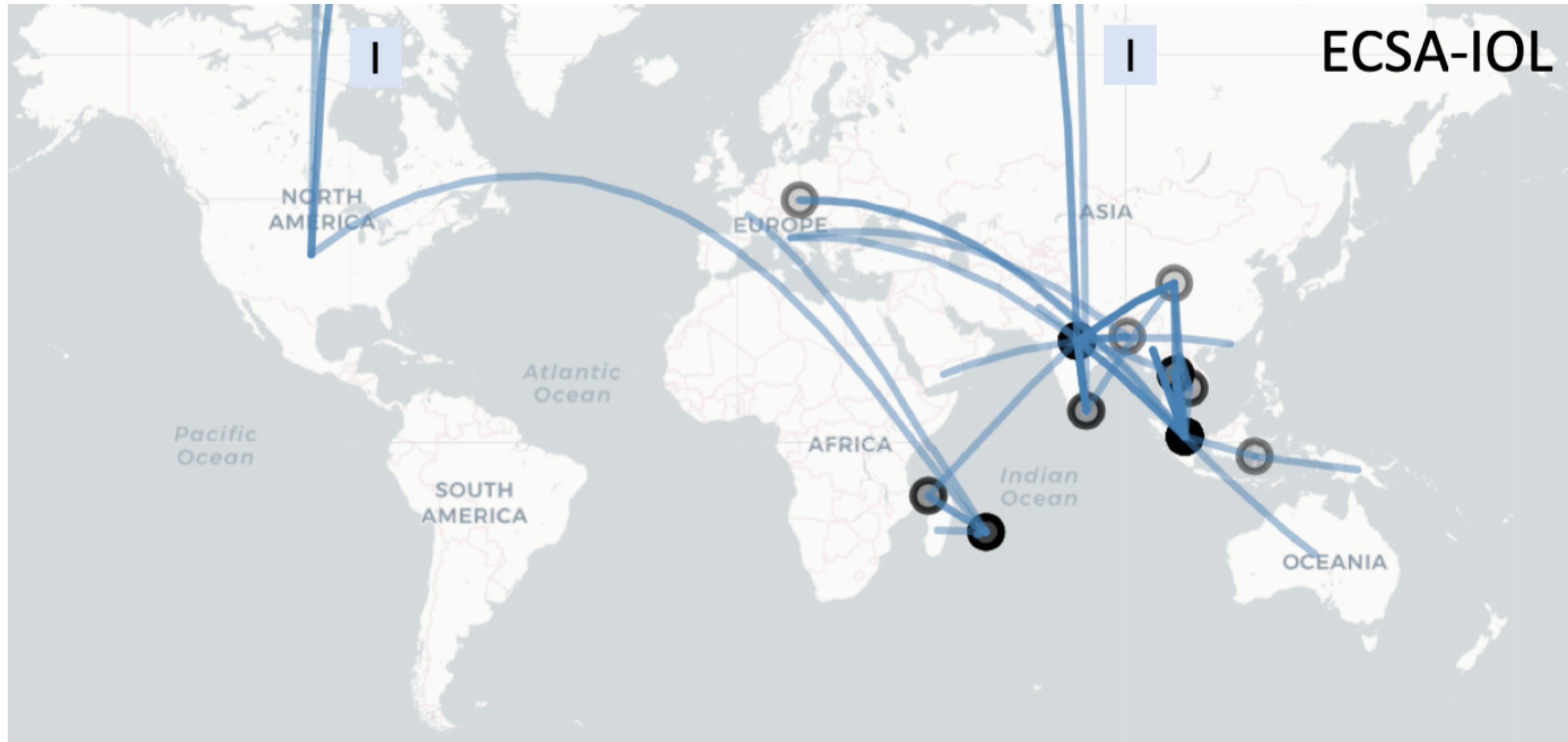
- 590 CHIKV genomes
- iq-tree / SH-aLRT



CHIKV Phylogeography AUL



CHIKV Phylogeography ECSA-IOL

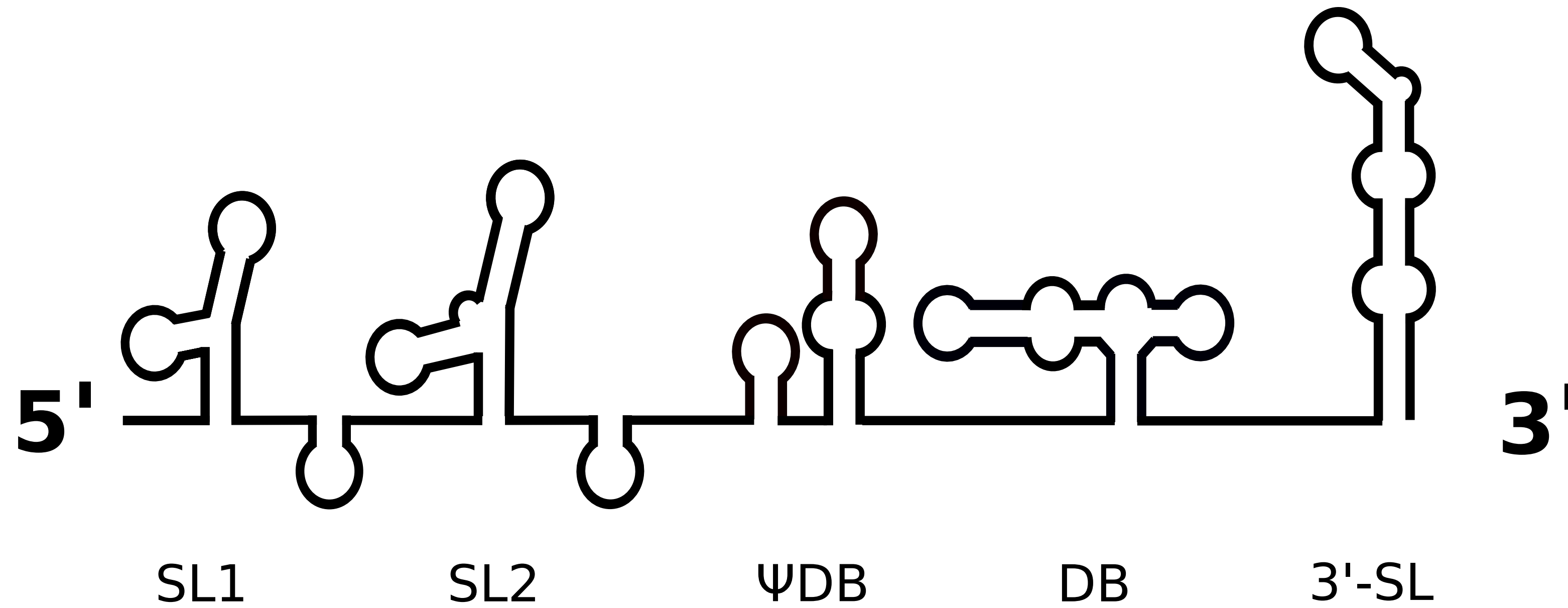


CHIKV Phylogeography ECSA-MASA

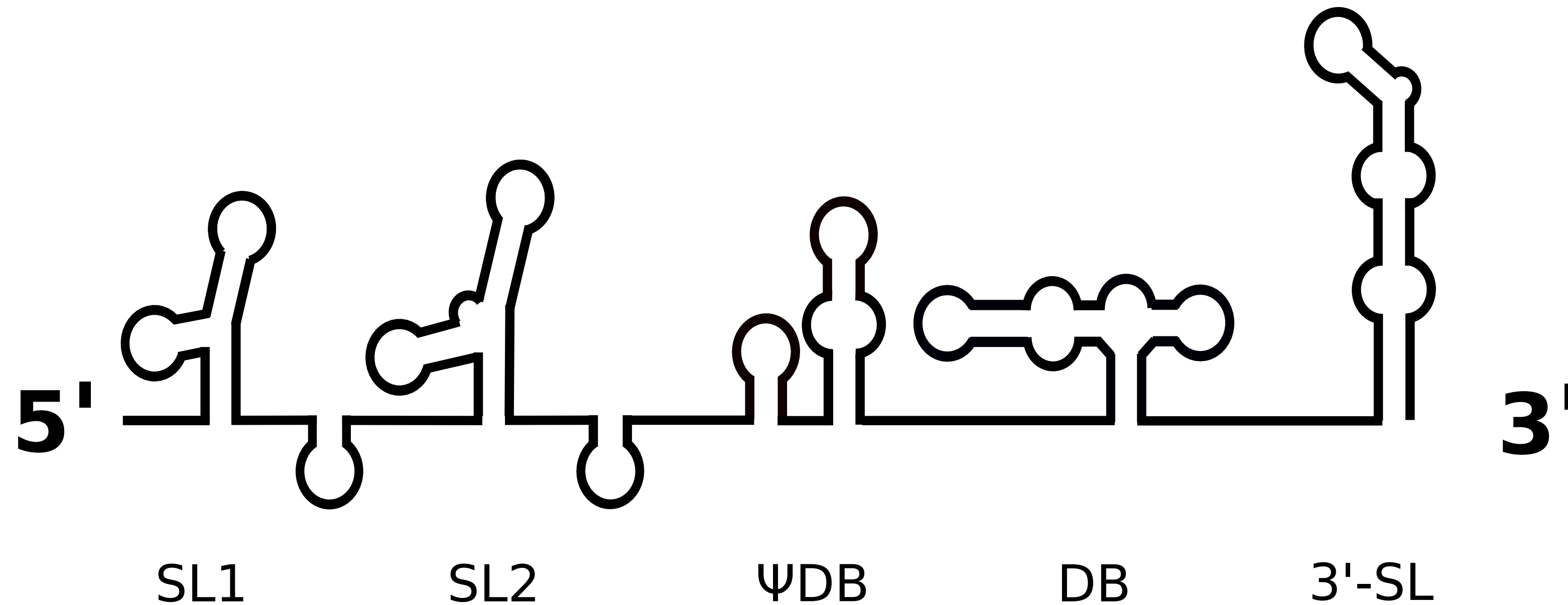


Part II: Conserved ncRNAs

Conserved RNAs in Viral 3'UTR: ZIKV

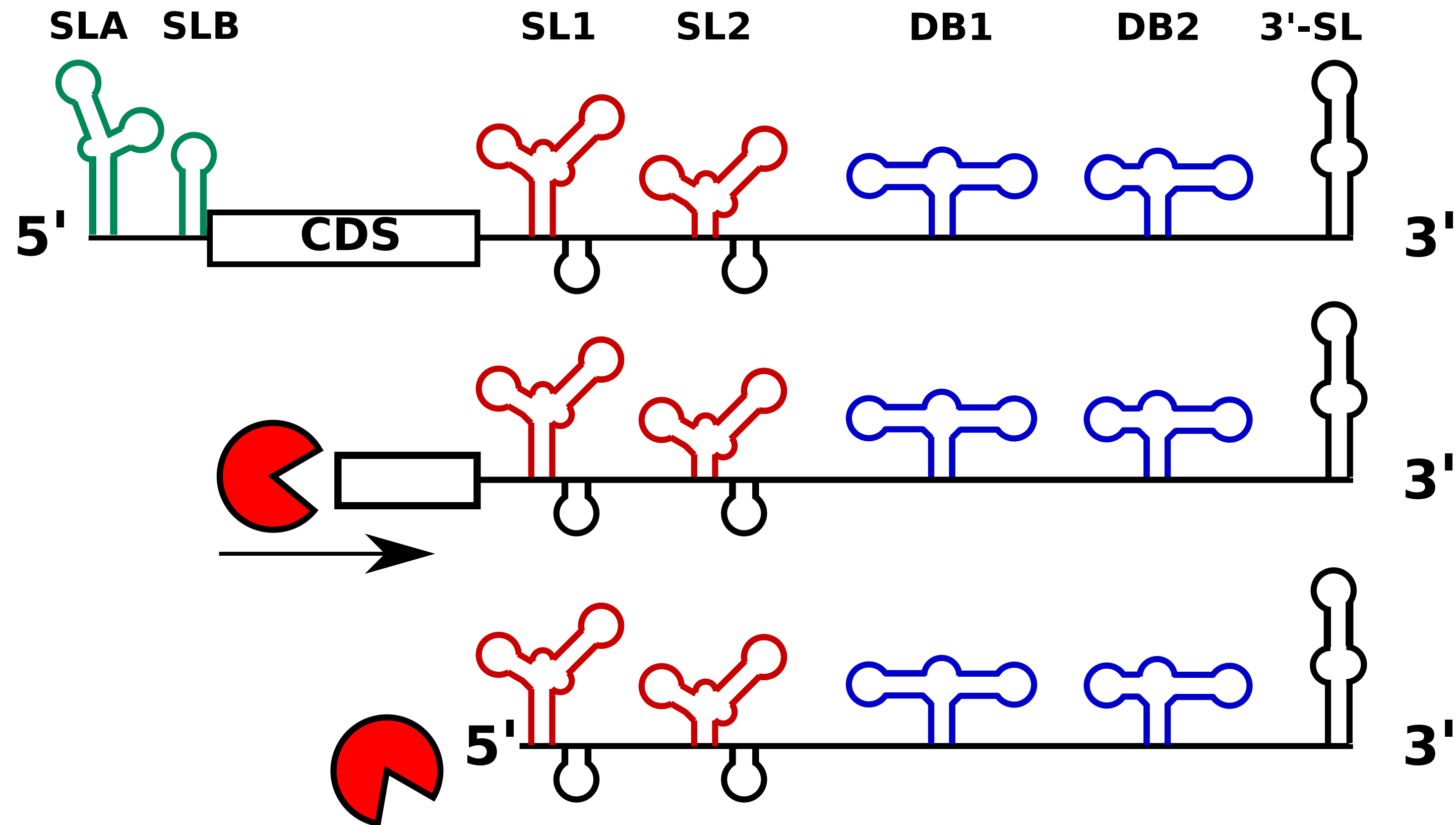


Conserved RNAs in Viral 3'UTR: ZIKV



- Accumulation of short flavivirus RNA (sfRNA) upon infection
- Stable decay intermediates produced by partial exoribonuclease degradation
- Xrn1 is efficiently stalled at conserved xrRNA structures

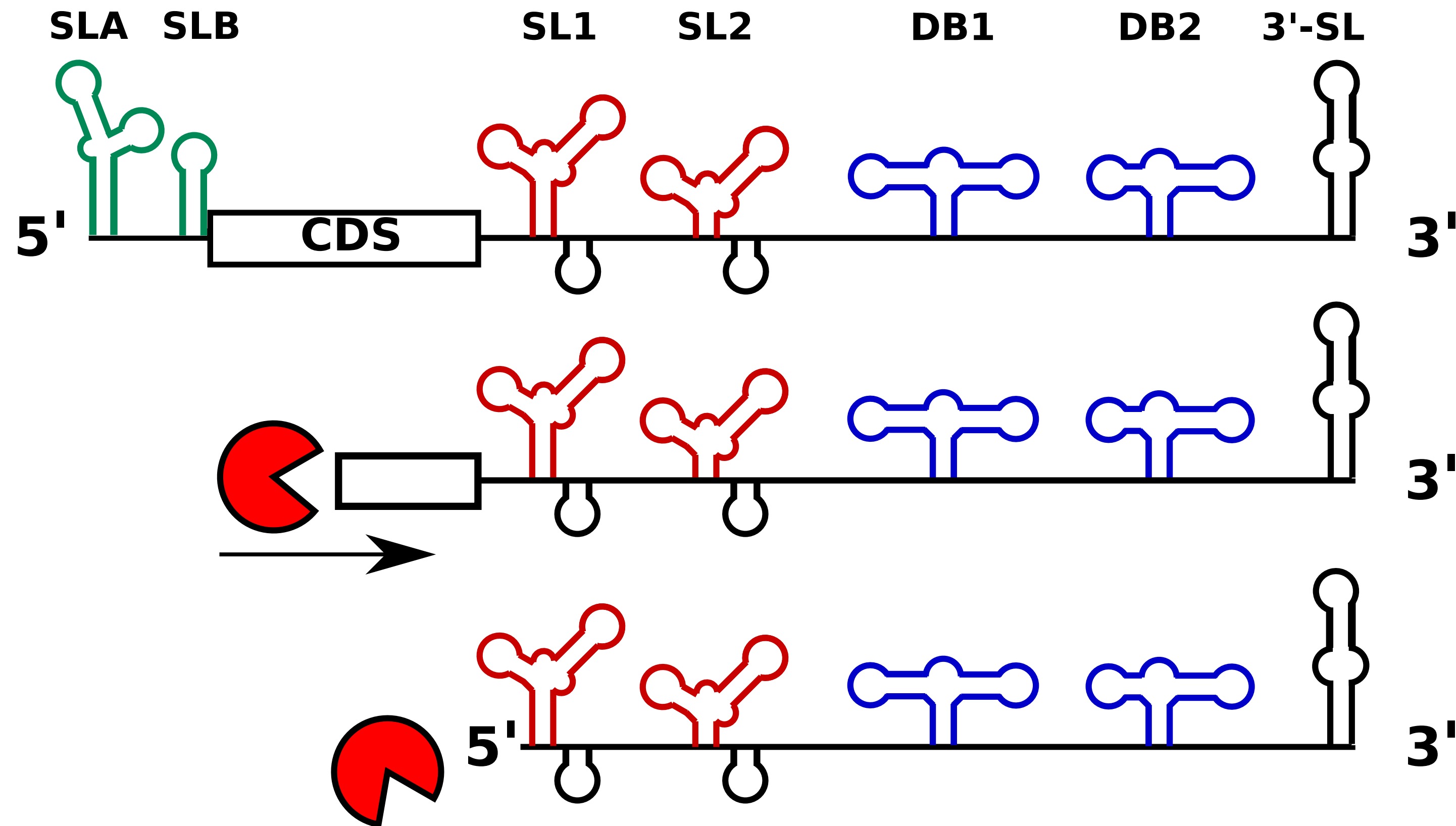
Conserved RNAs in Viral 3'UTR



EXCURSION

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Conserved RNAs in Viral 3'UTR

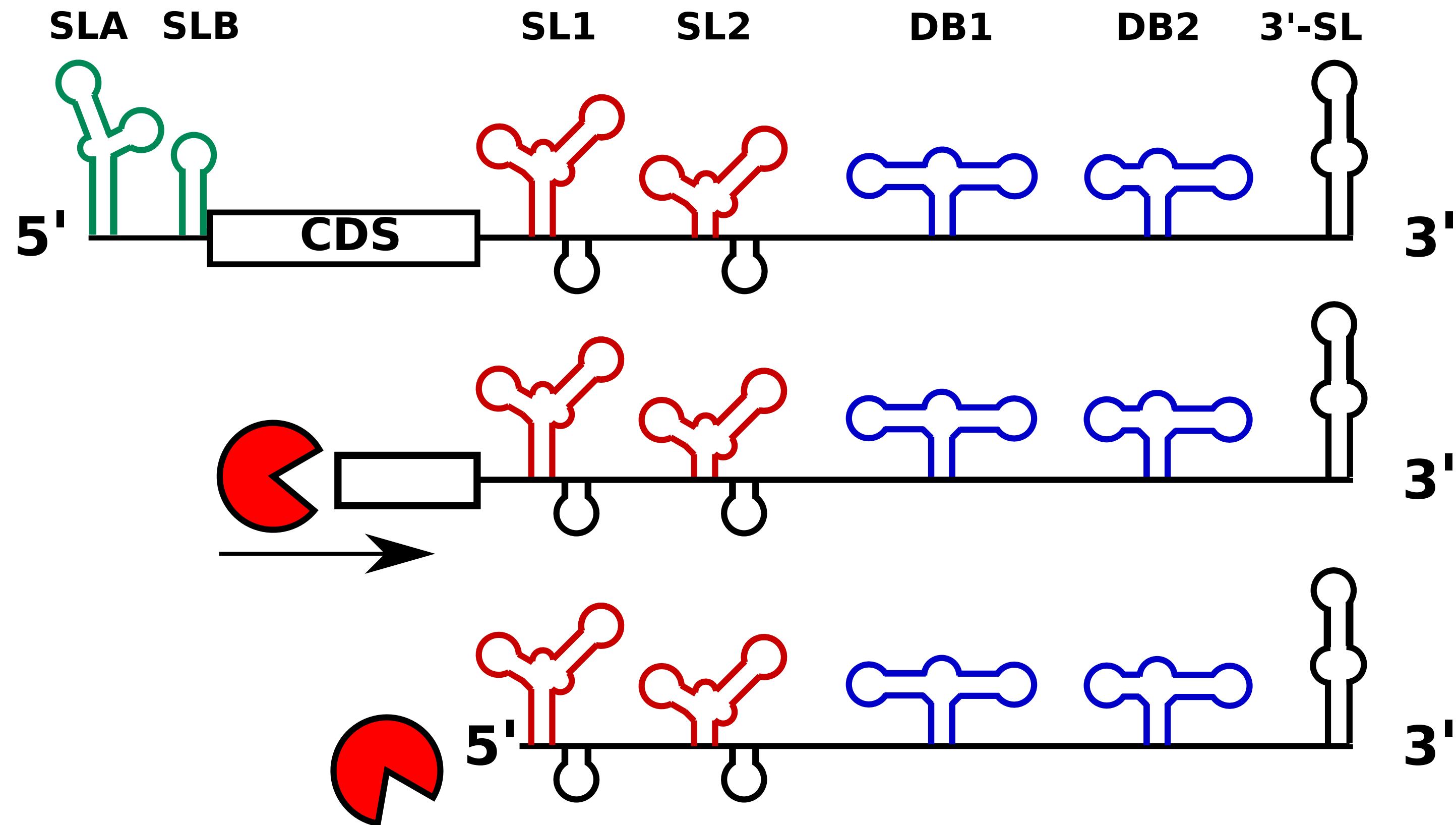


**sfRNA mediates
pathogenicity !**

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Conserved RNAs in Viral 3'UTR

EXCURSION



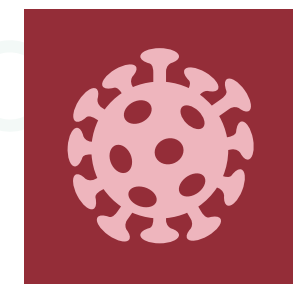
**sfRNA mediates
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How to find them?

- Accumulation of short flavivirus RNA (sfRNA) upon infection
- Stable decay intermediates produced by partial exoribonuclease degradation
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Conserved RNAs in Viral 3'UTR

EXCURSION



viruses

<https://doi.org/10.3390/v11030298>



Article

Functional RNA Structures in the 3'UTR of Tick-Borne, Insect-Specific and No-Known-Vector Flaviviruses

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² Research Group BCB, Faculty of Computer Science, University of Vienna, Währingerstraße 29, 1090 Vienna, Austria

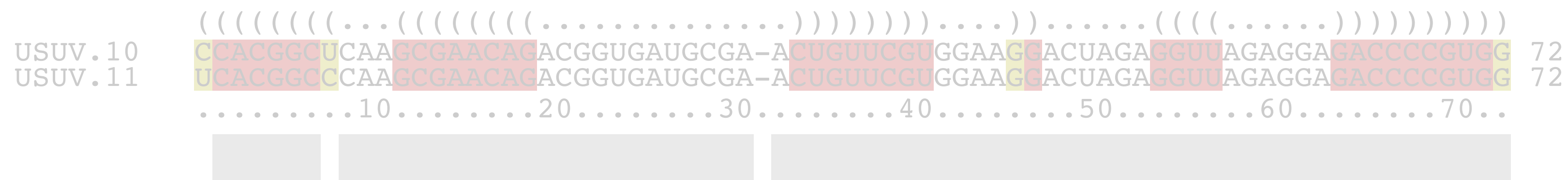
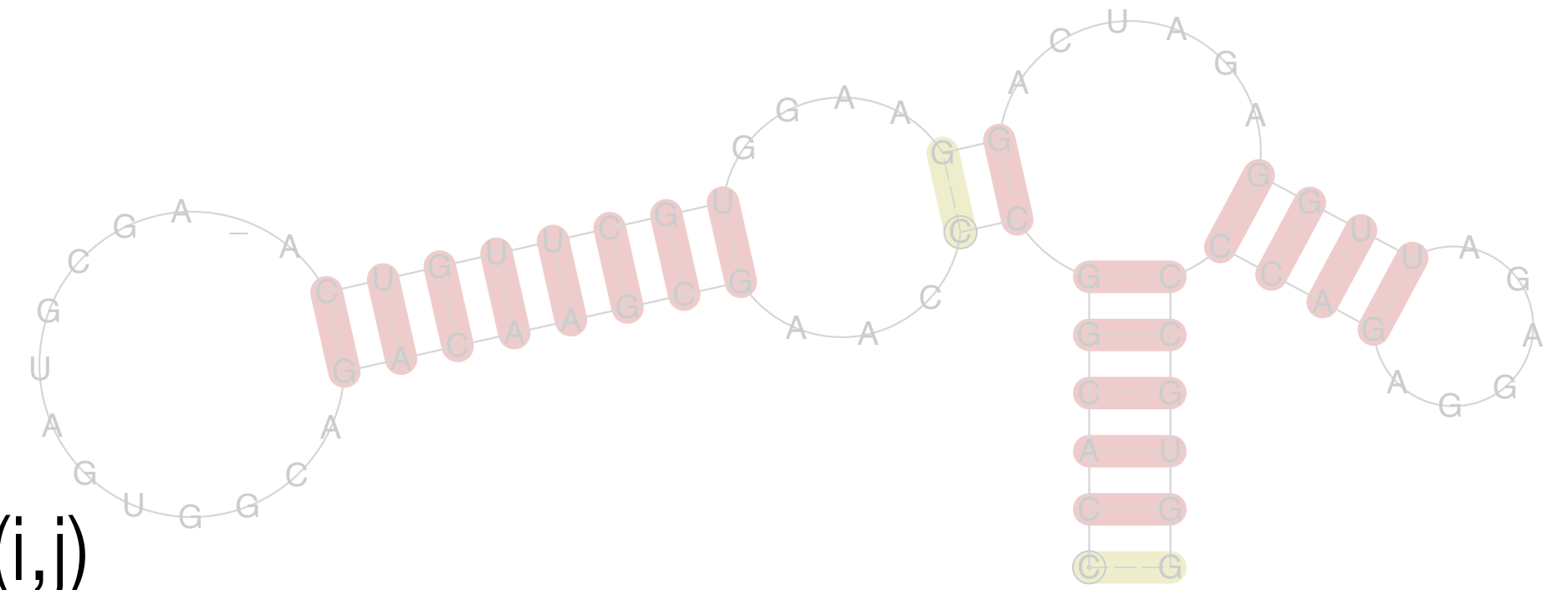
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RNA Covariation as Evolutionary Trait

- High mutation rate in RNA viruses due to error-prone RdRP
- For base pair (i,j): GC/CG/AU/UA/GU/UG
- Consistent mutation: different standard combinations
- Compensatory mutation: both positions are mutated
- Presence of both strongly supports predicted base pair (i,j)



types of pairs:

	1	2	3	4	5	6
0						
1						
2						

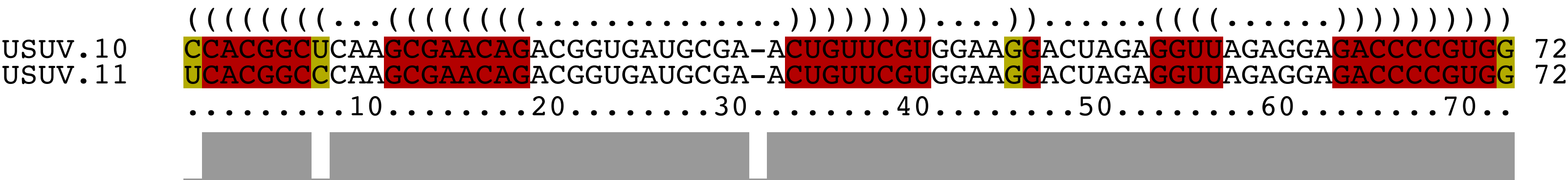
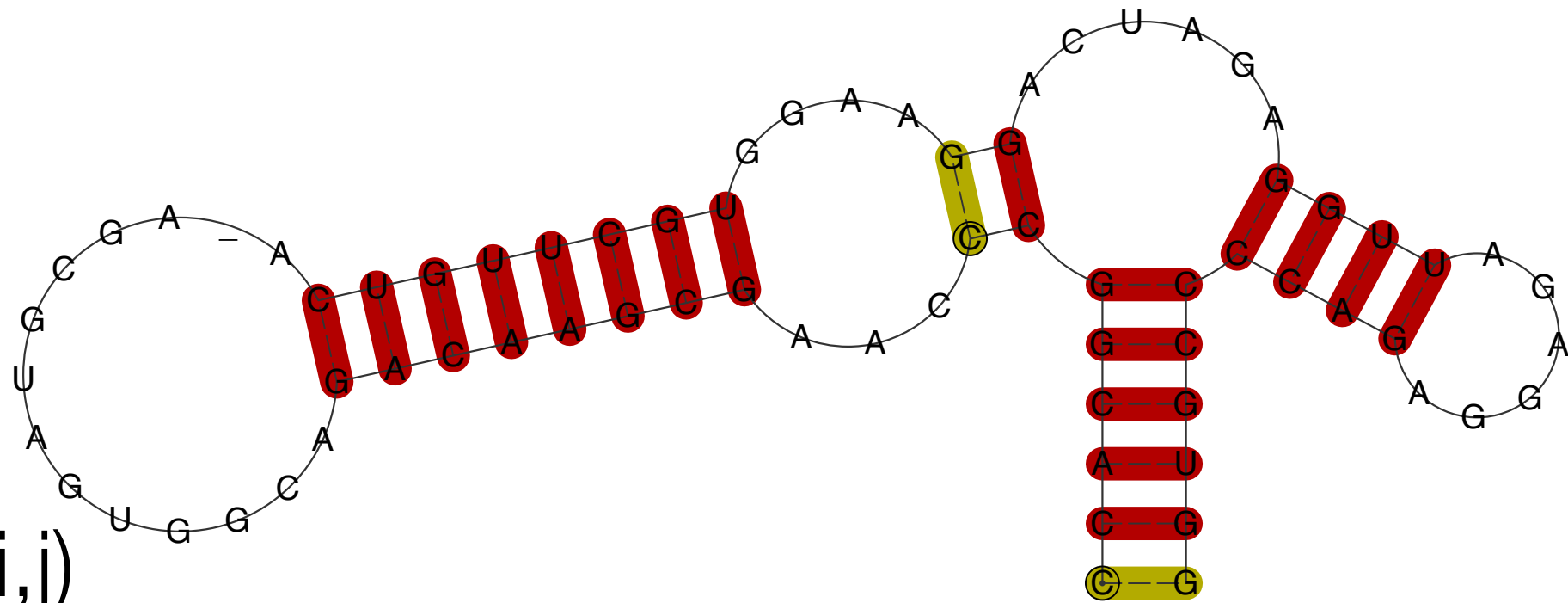
incompatible pairs in column

allowed pairs:

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U-A	C-G	U-G

RNA Covariation as Evolutionary Trait

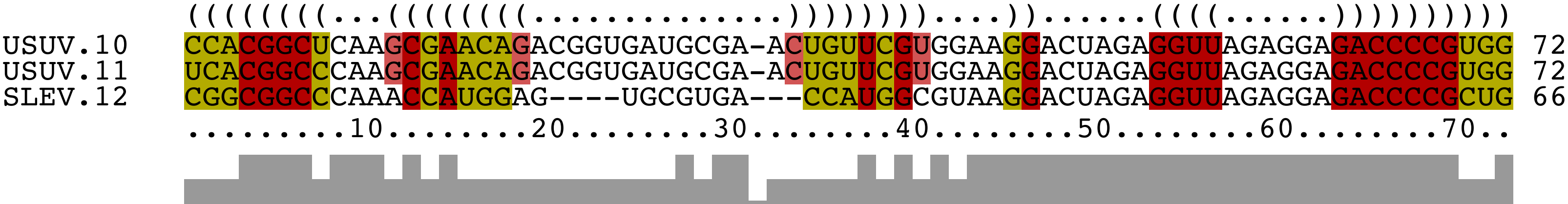
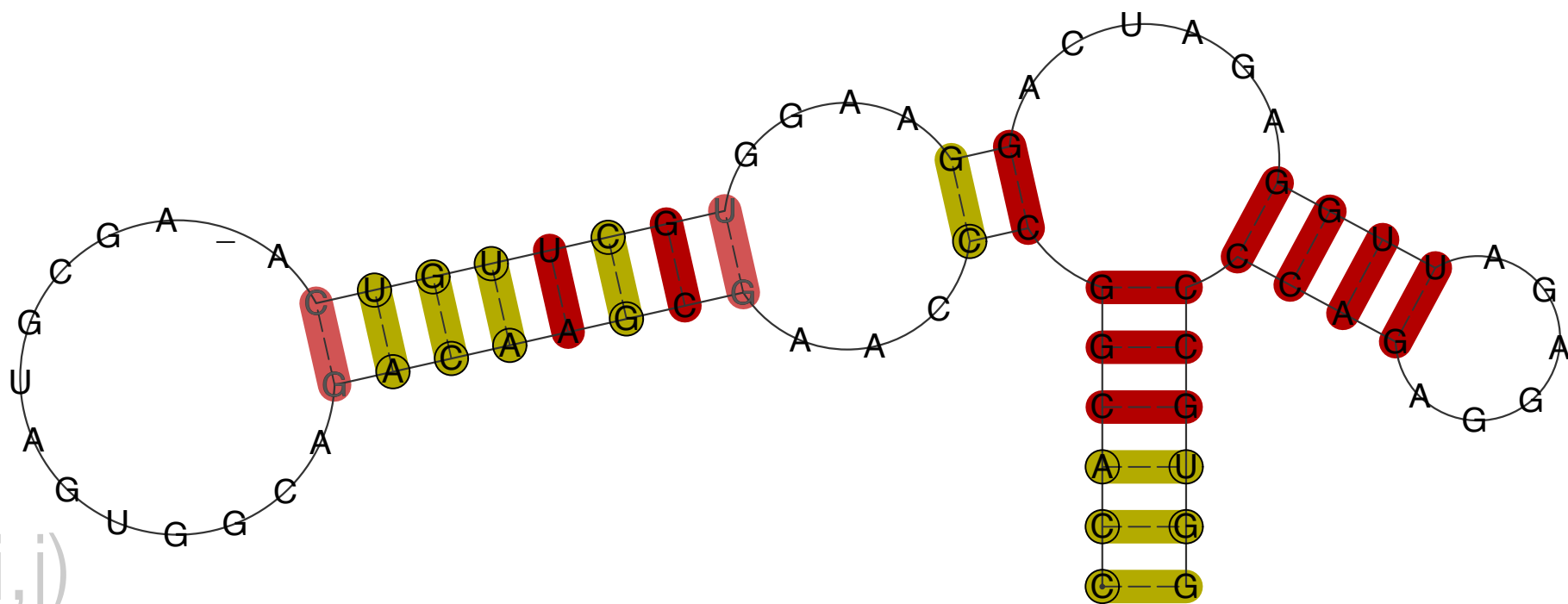
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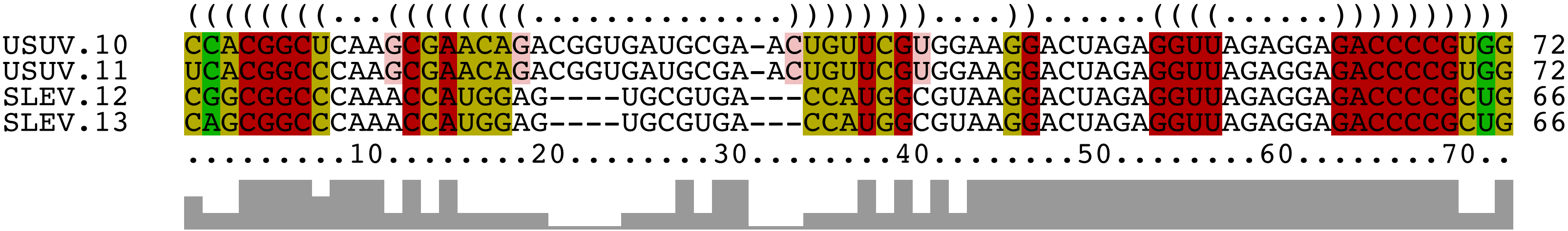
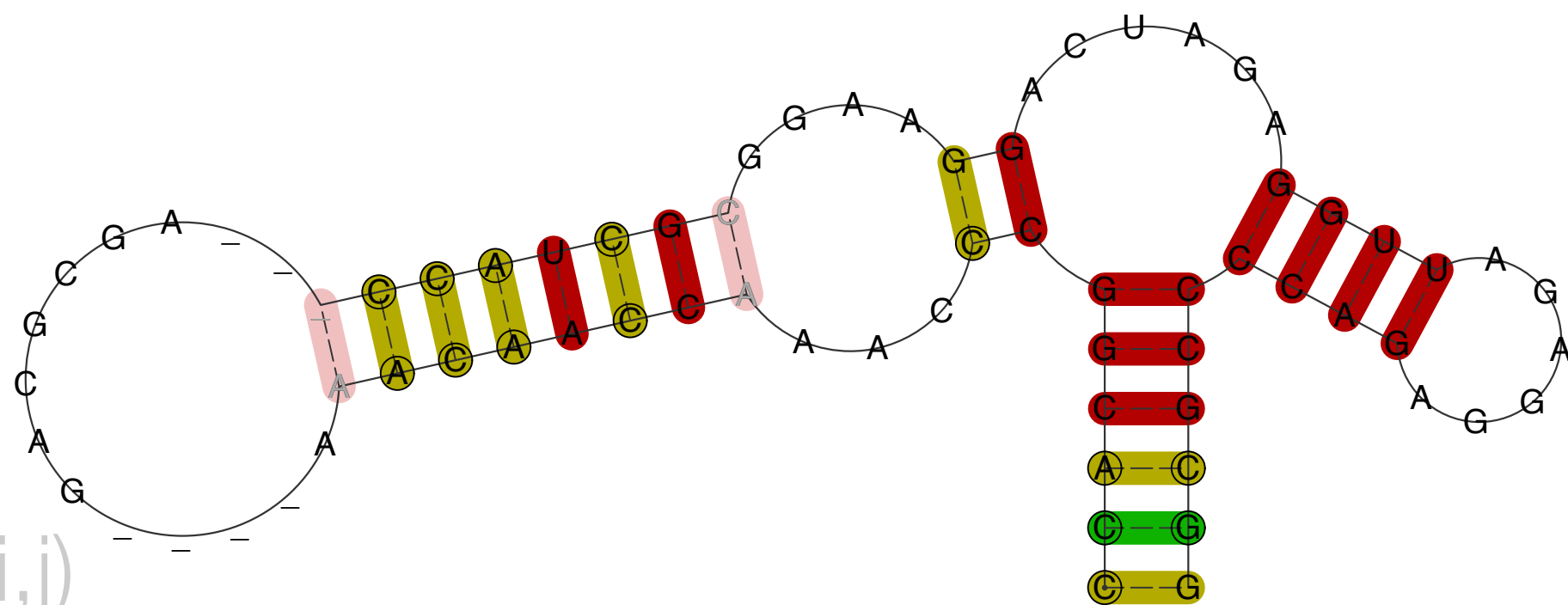
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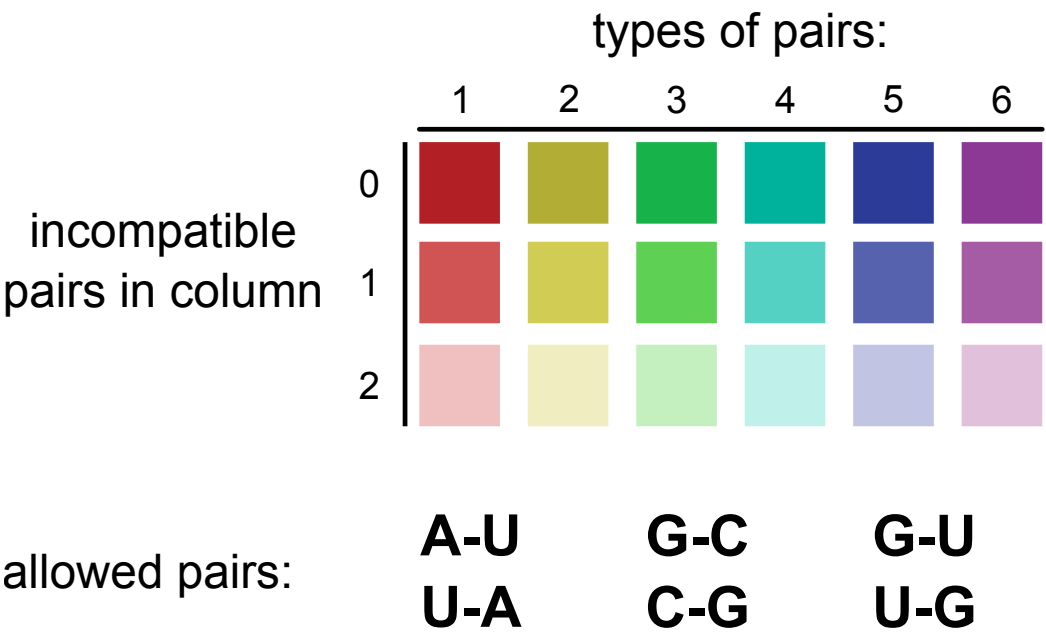
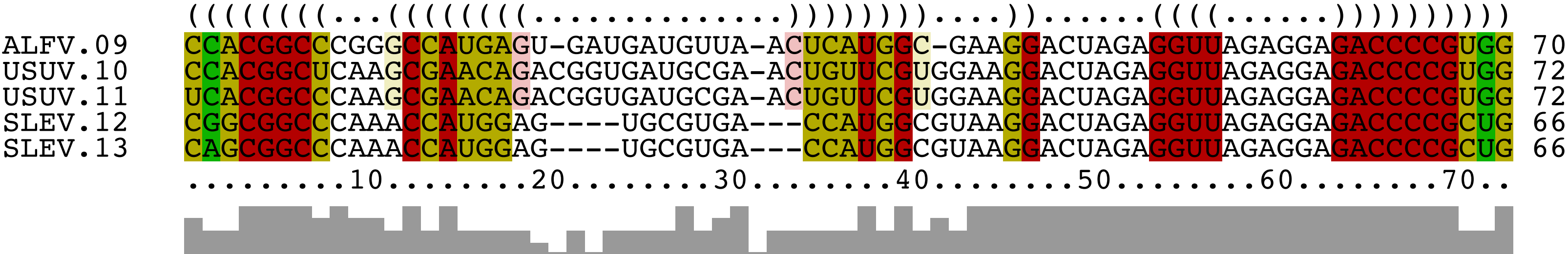
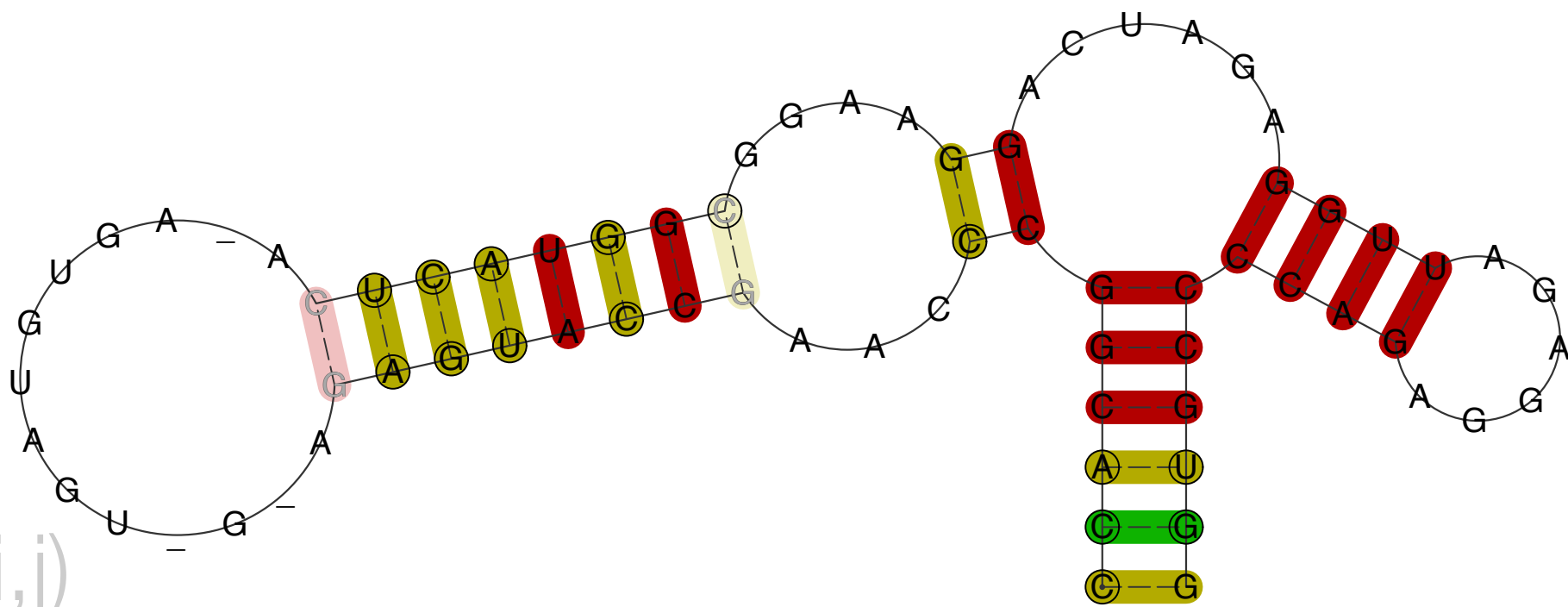
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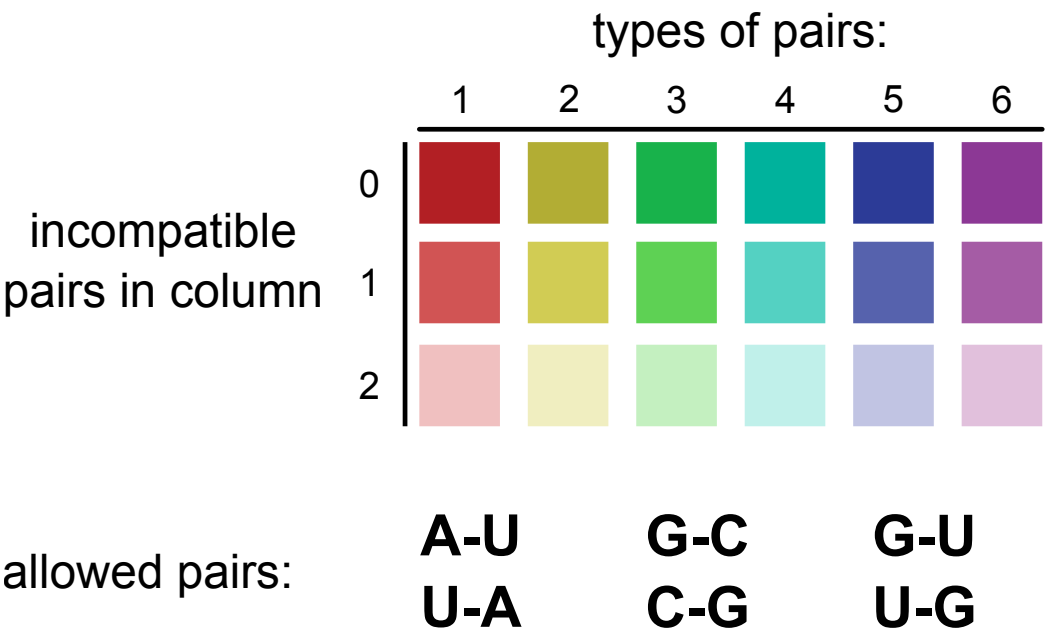
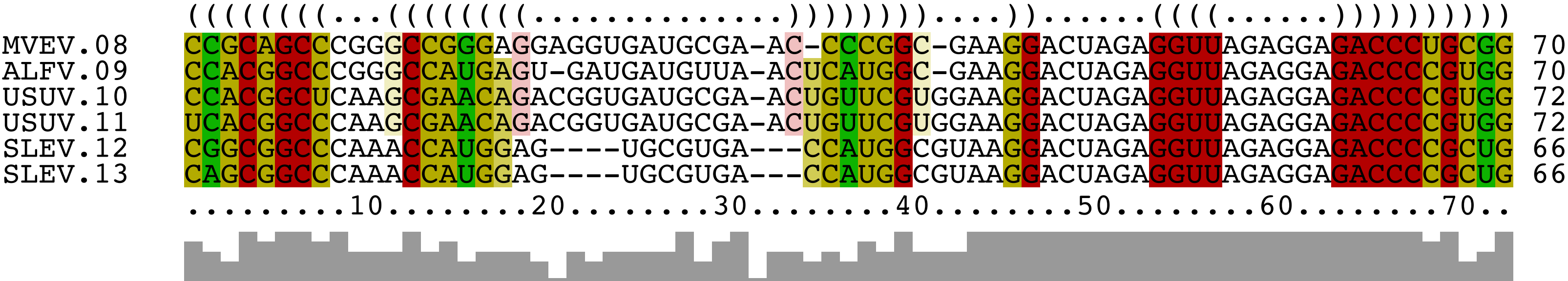
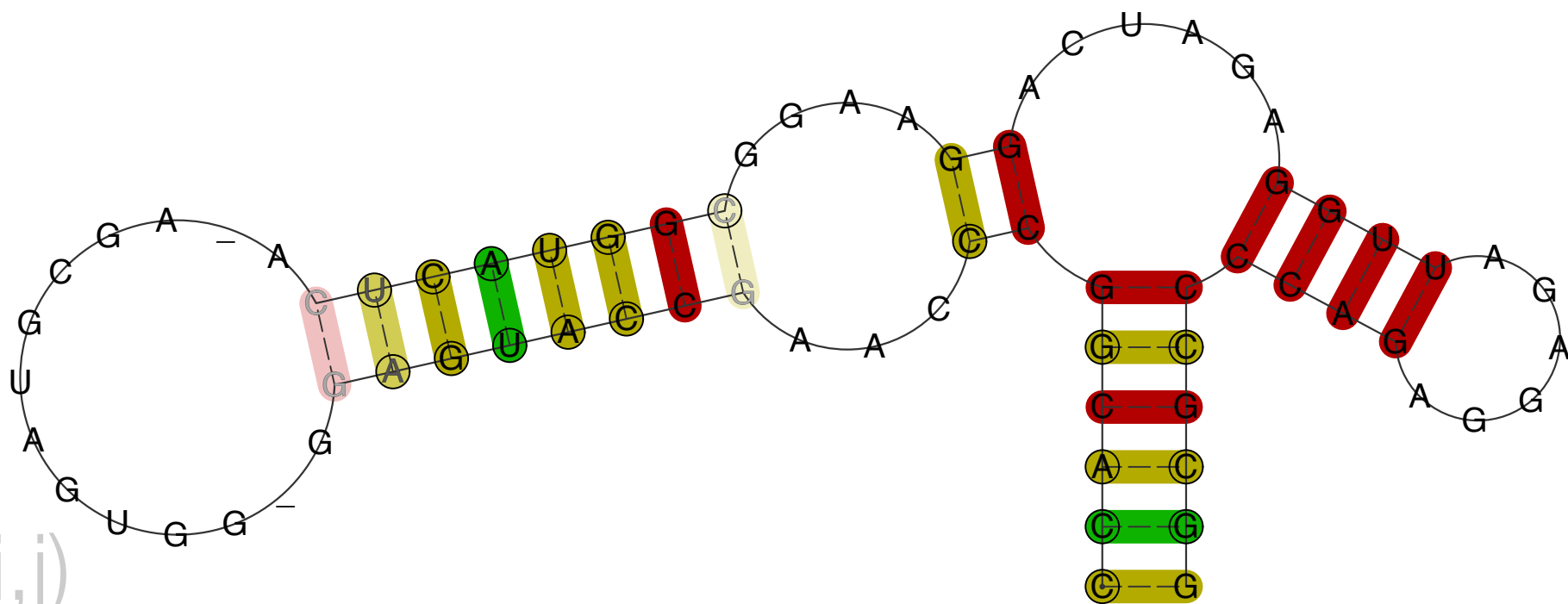
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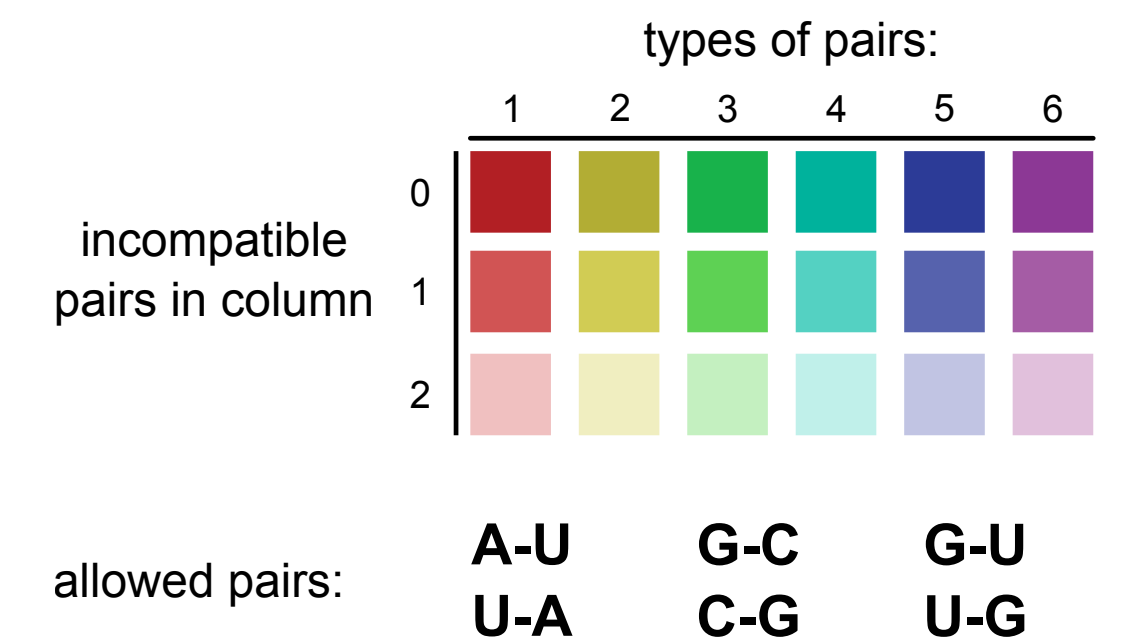
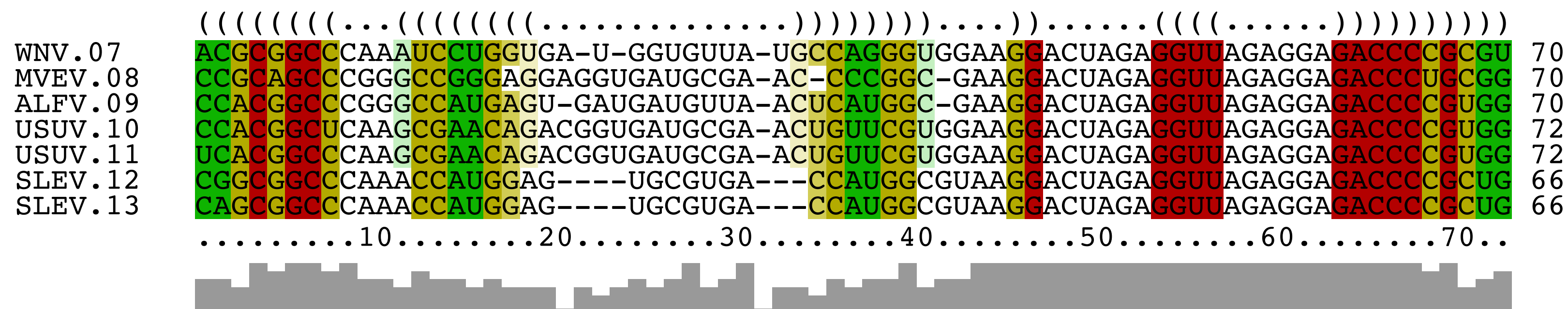
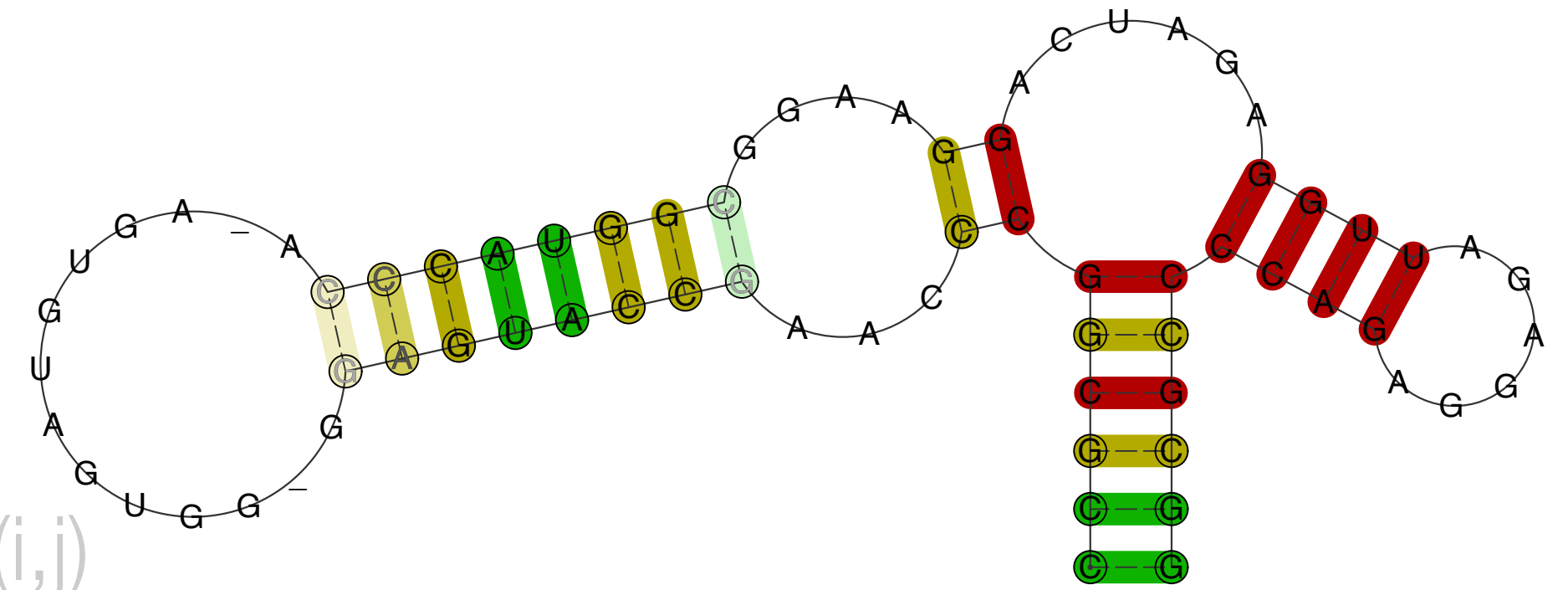


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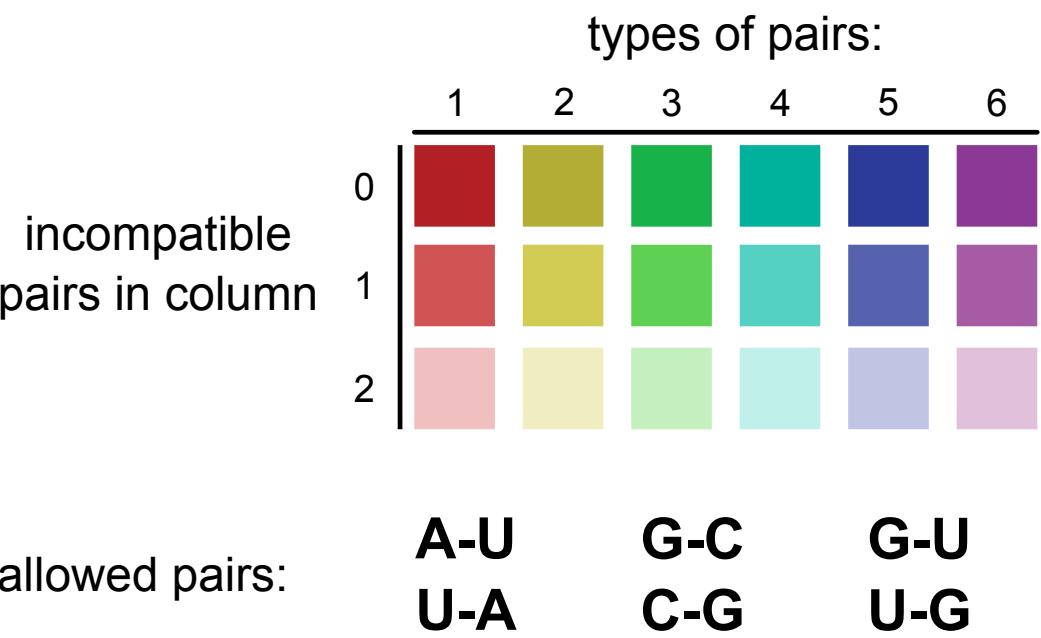
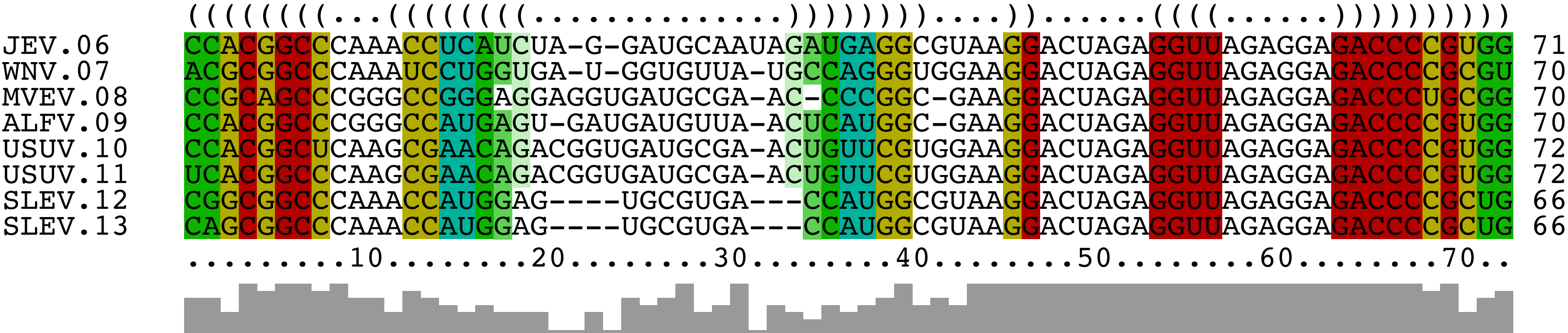
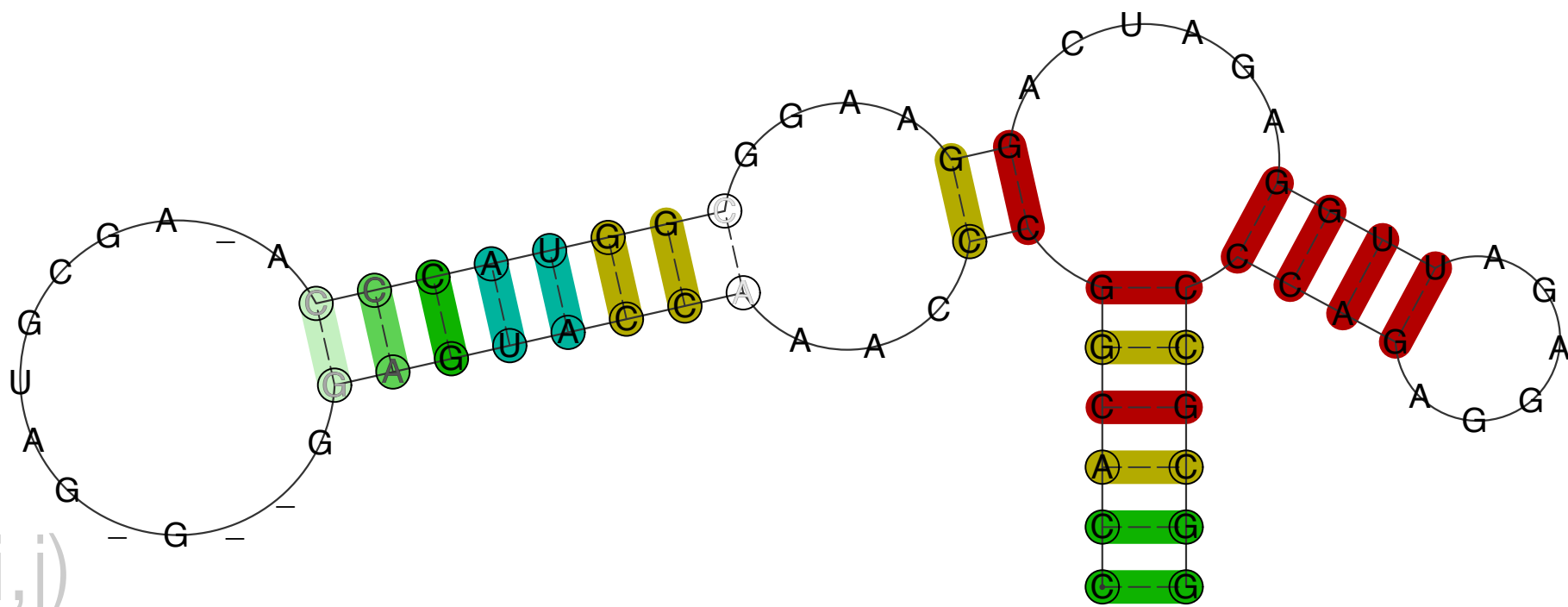


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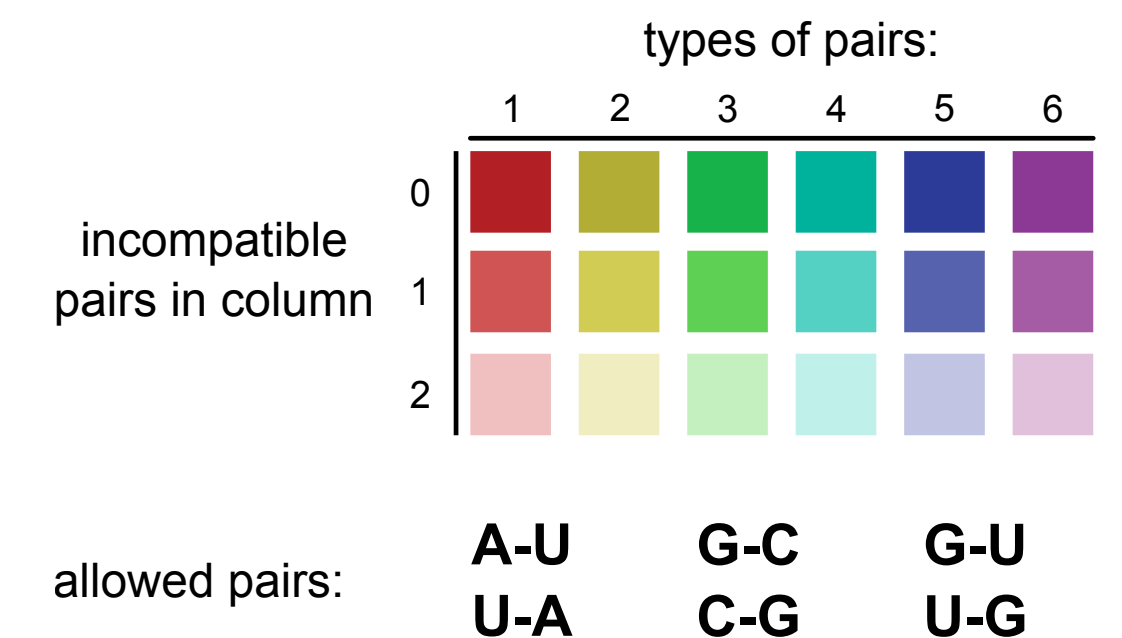
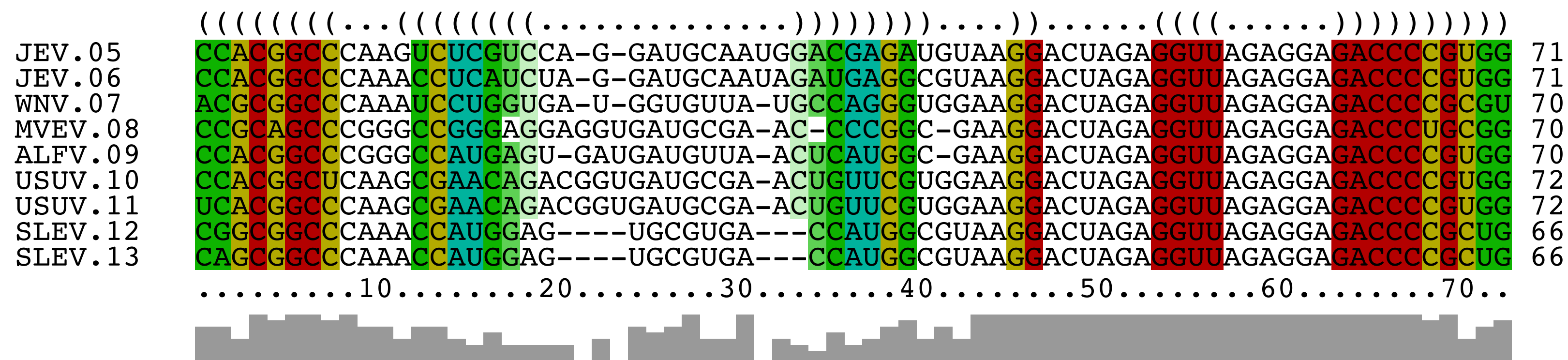
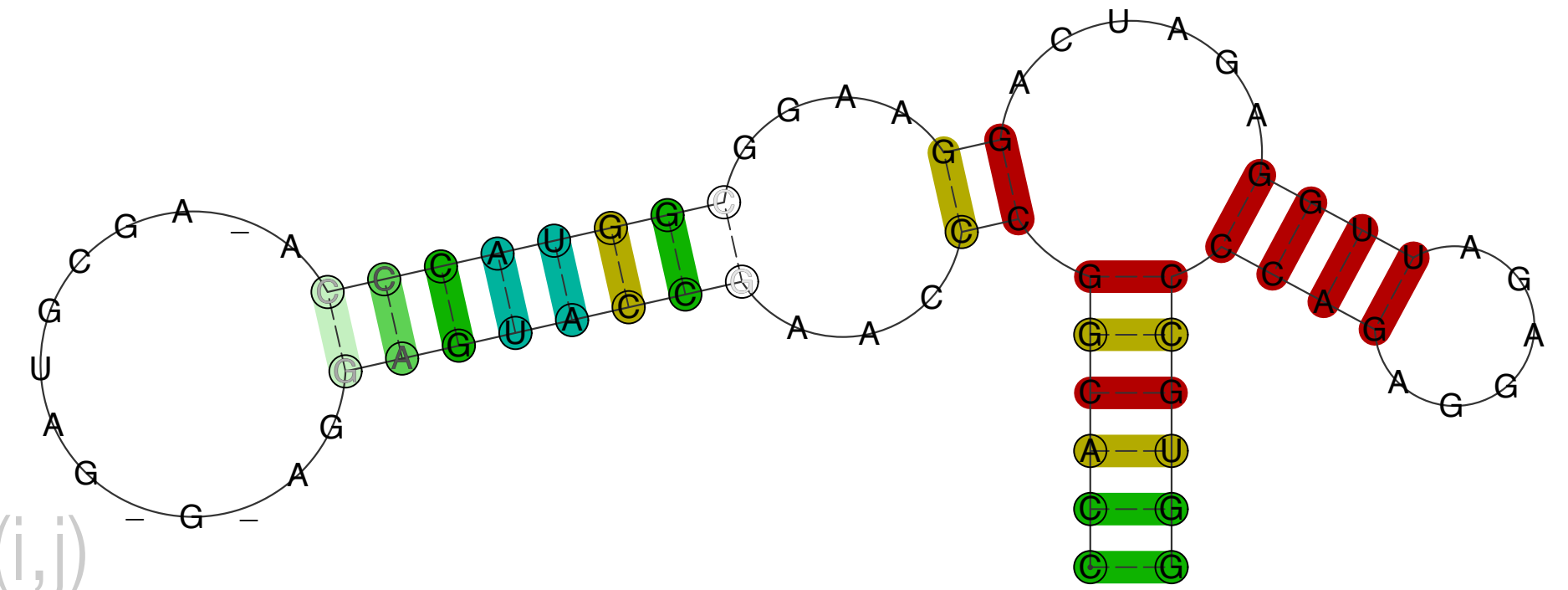


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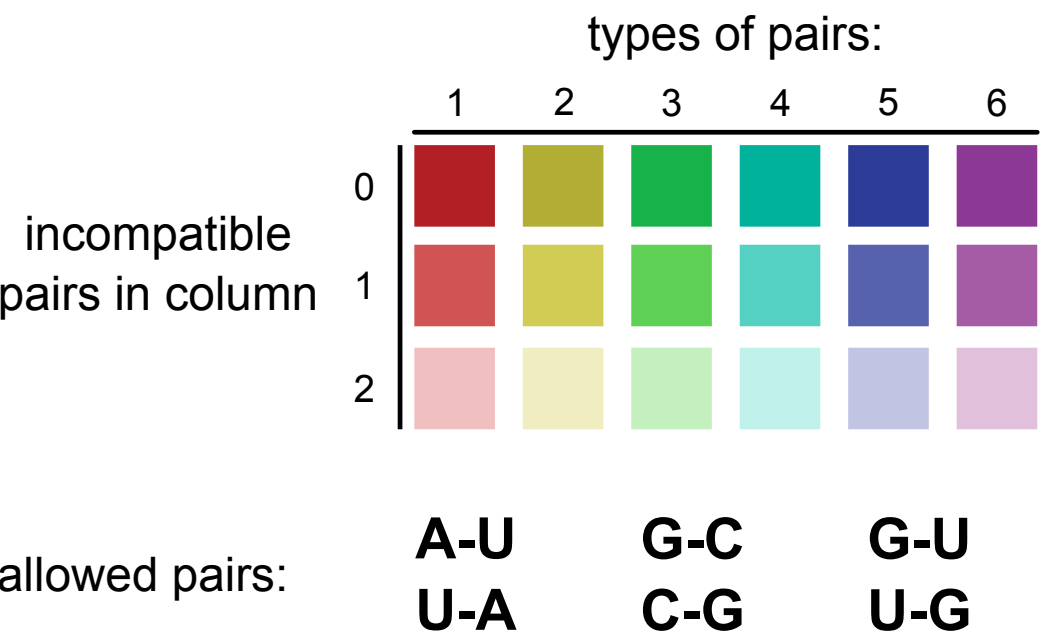
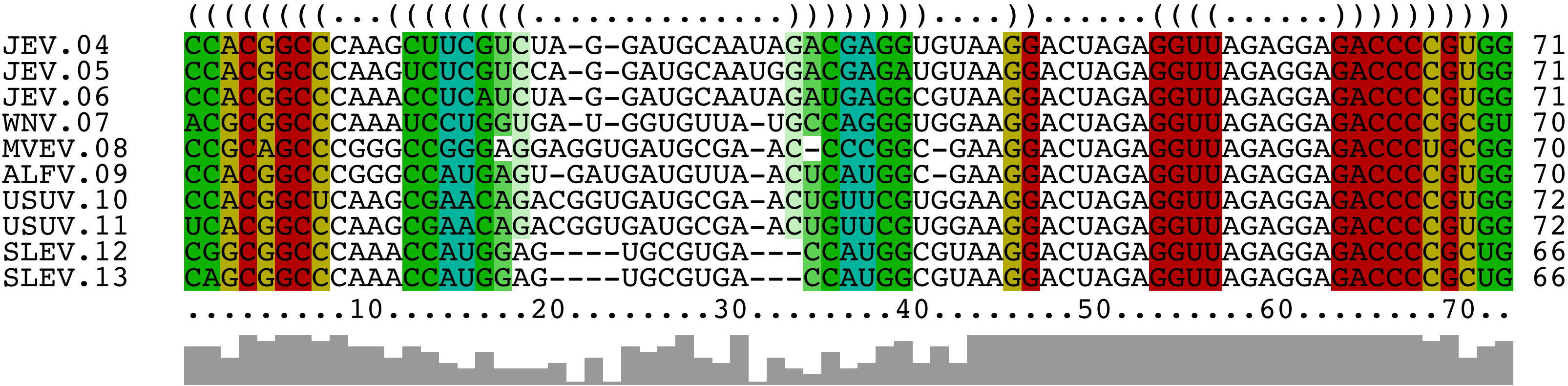
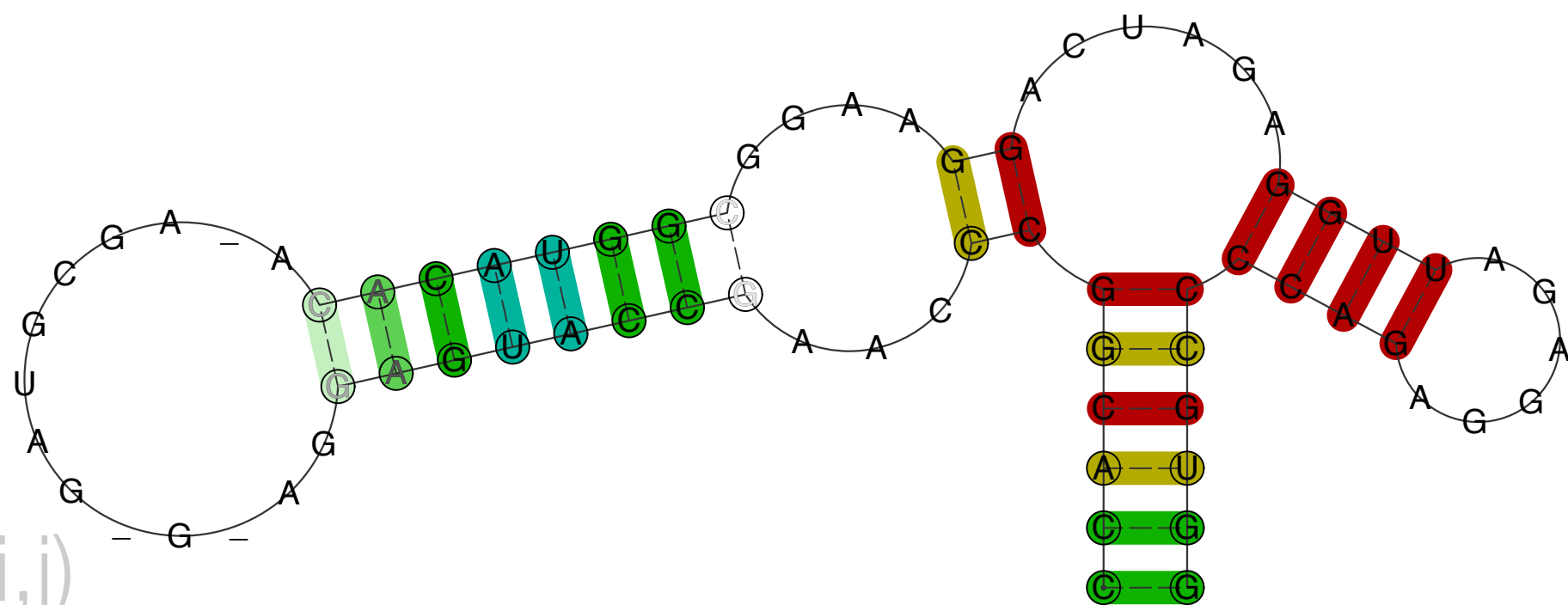


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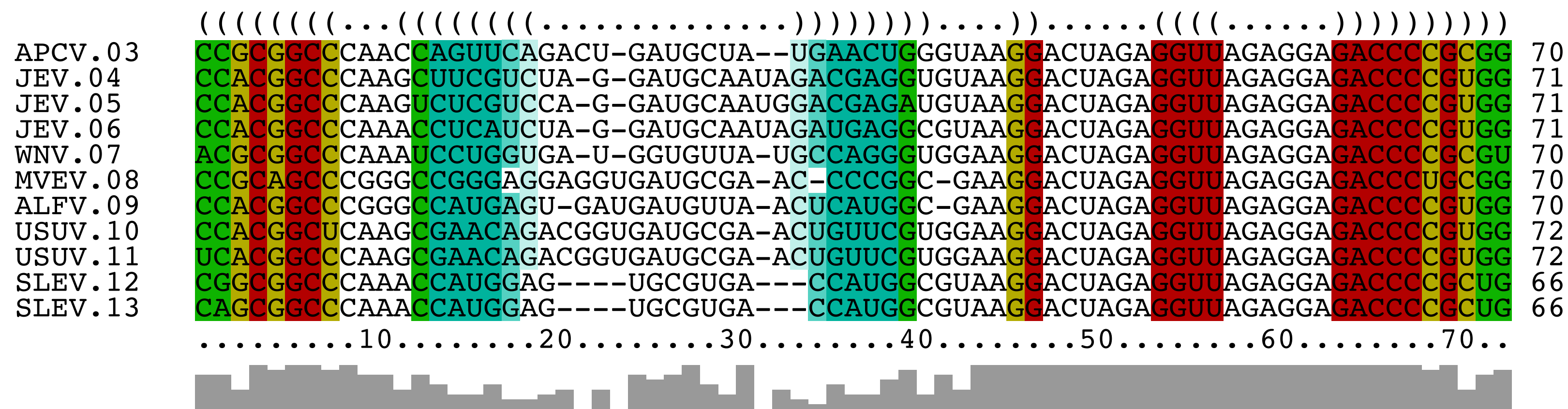
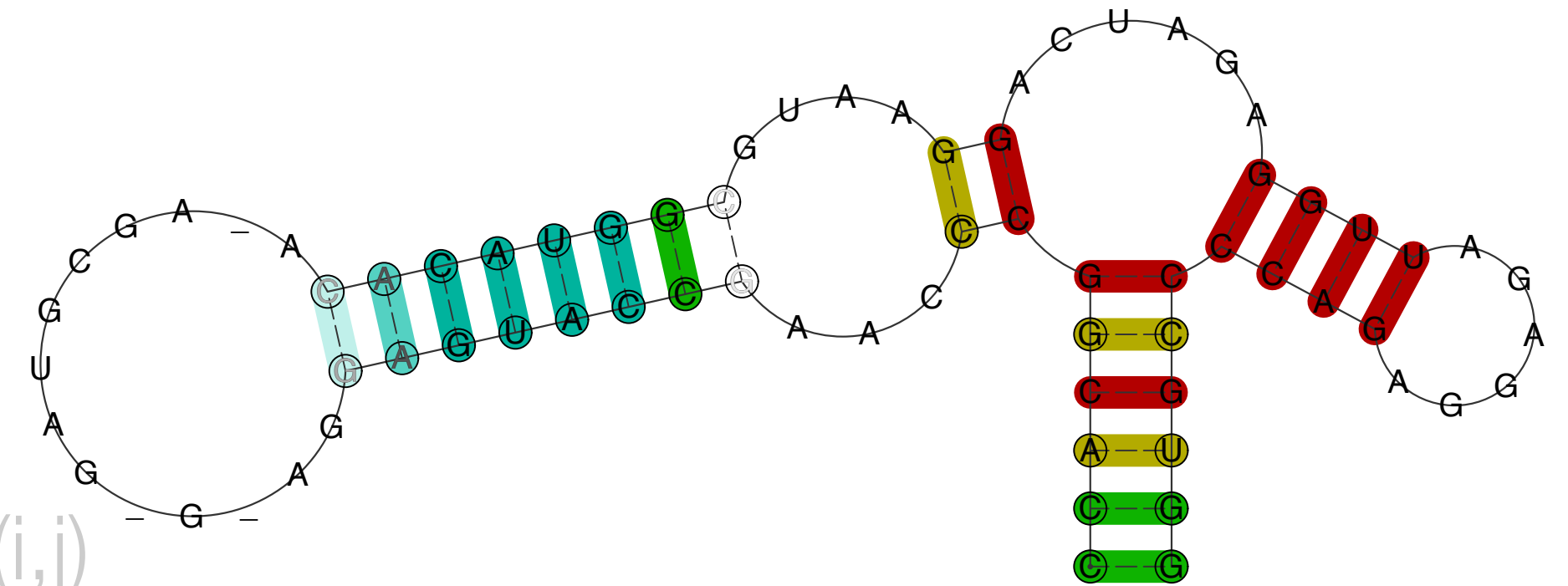
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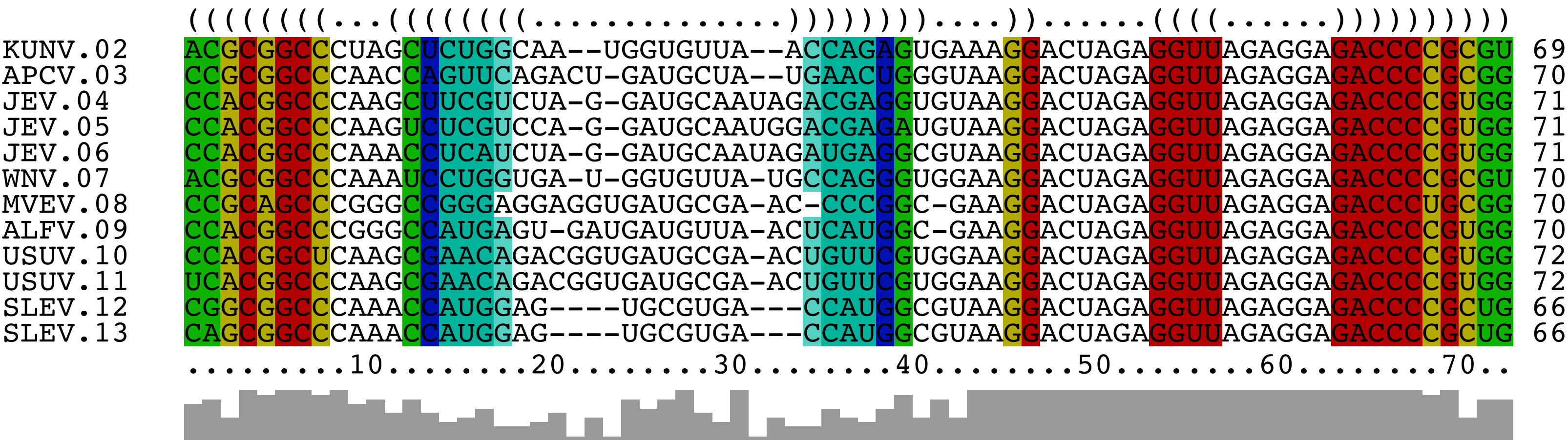
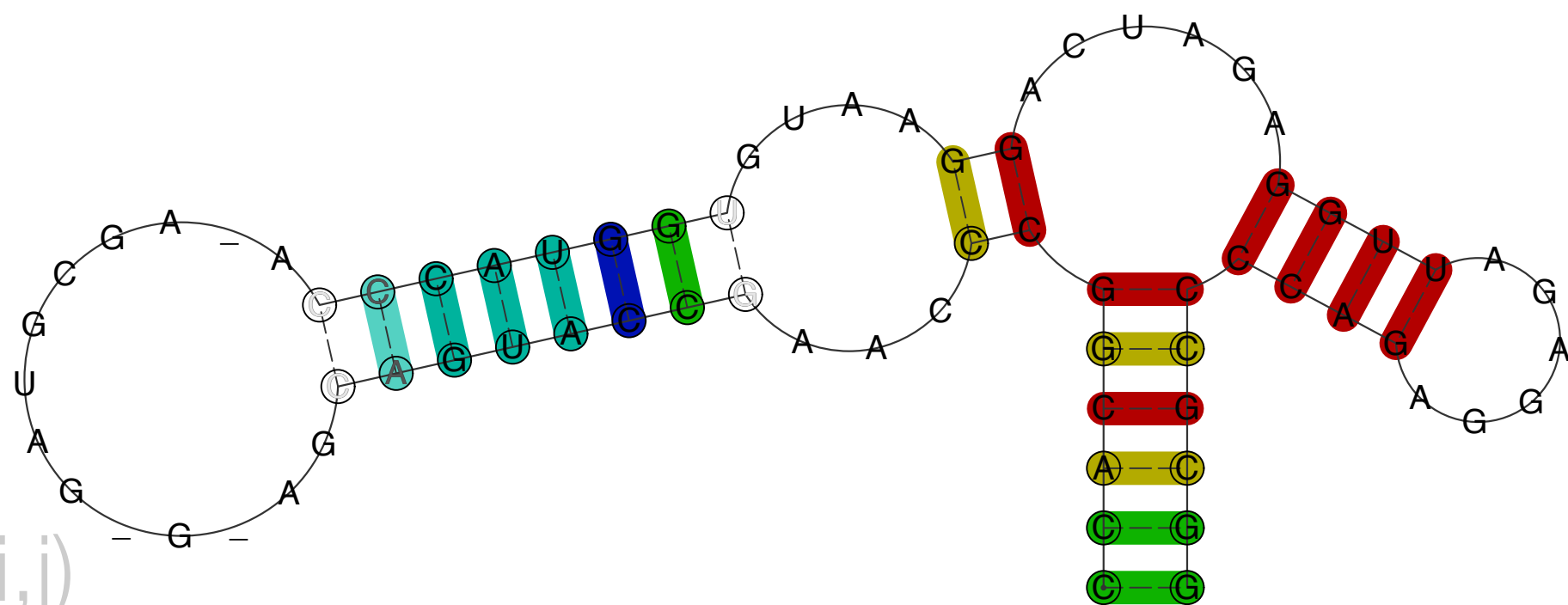
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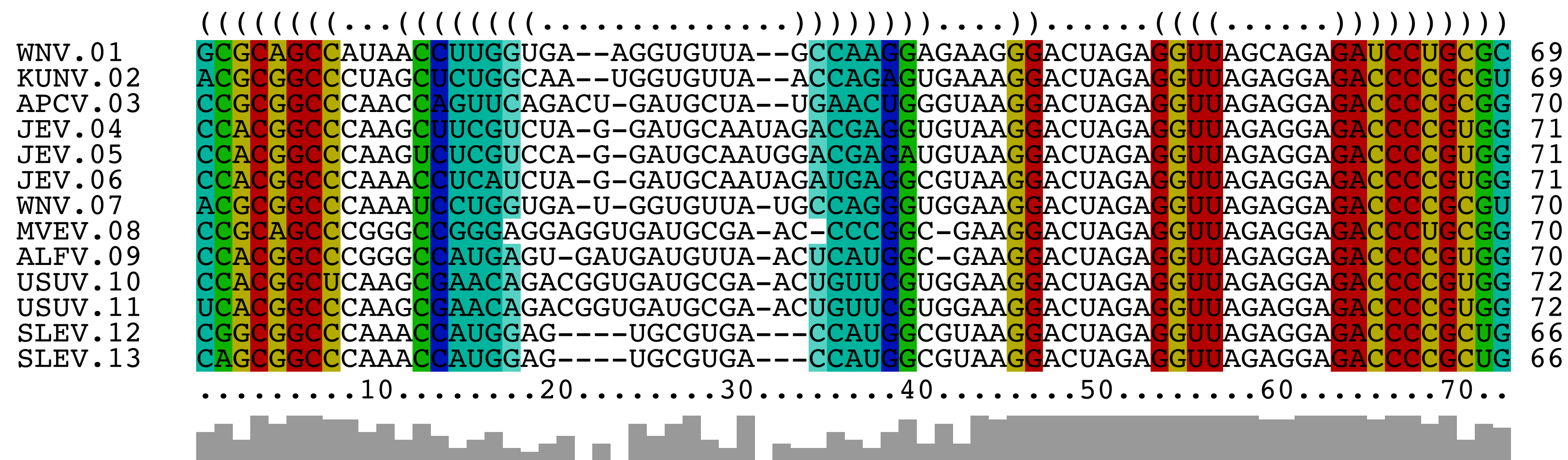
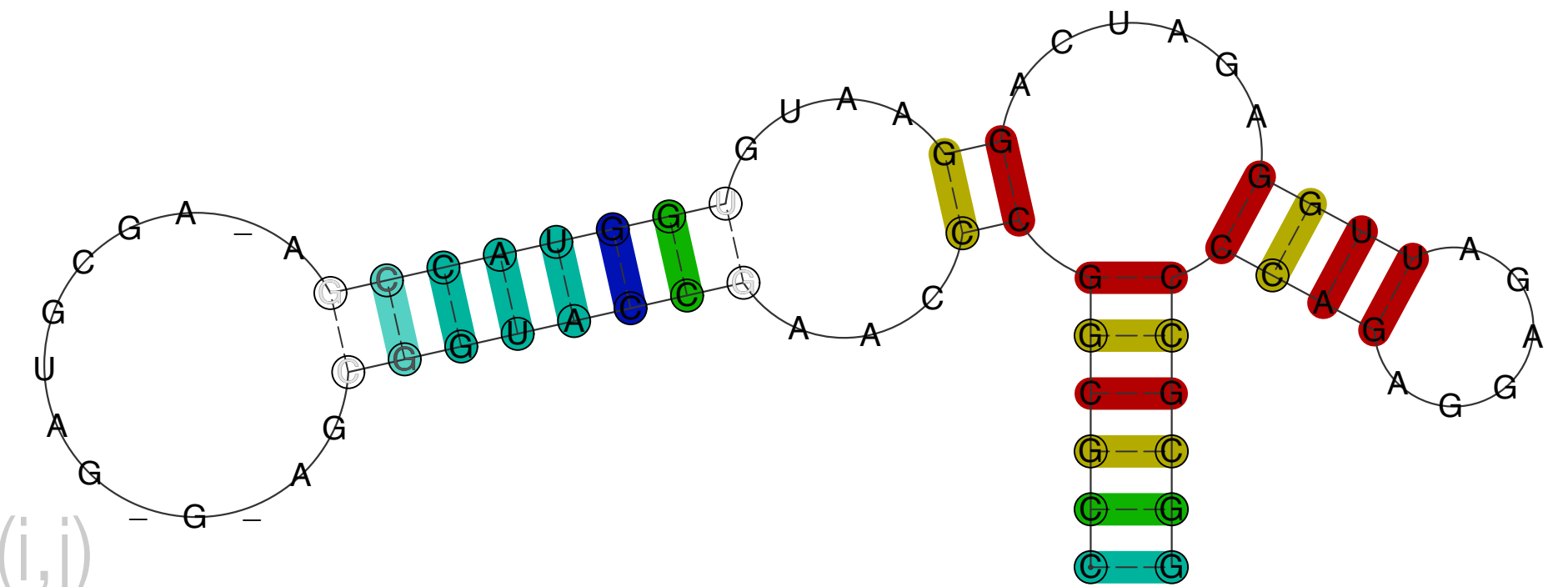
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RNA Covariation as Evolutionary Trait



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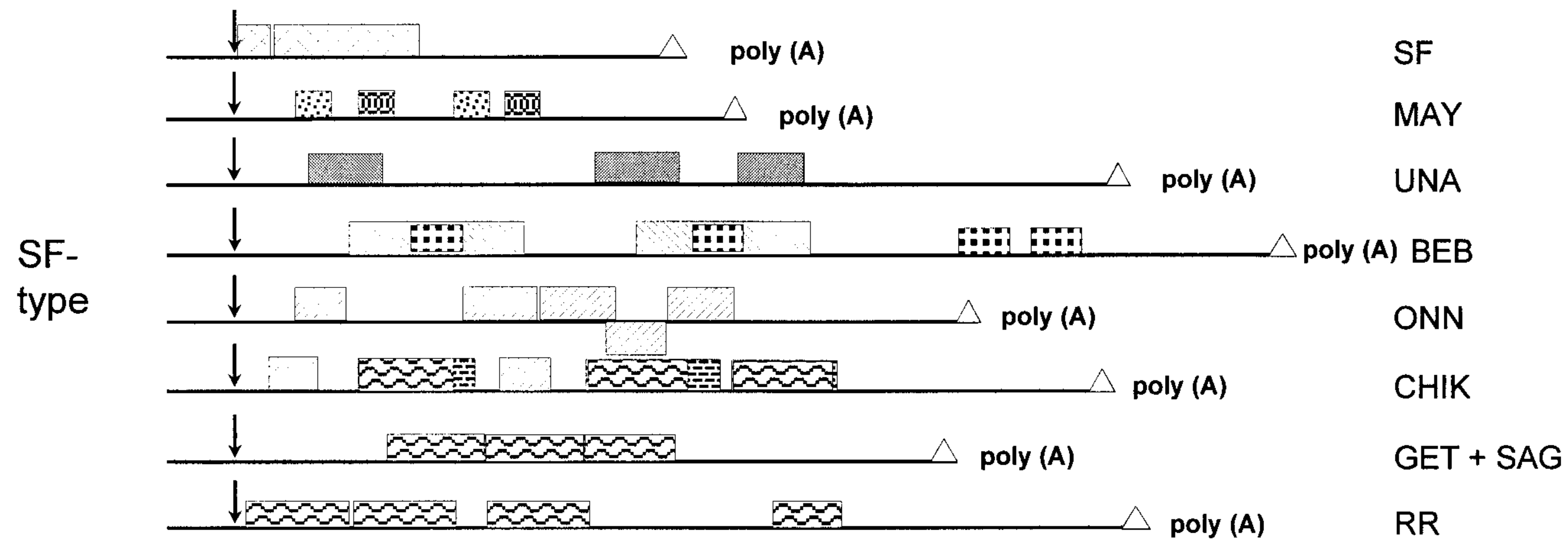
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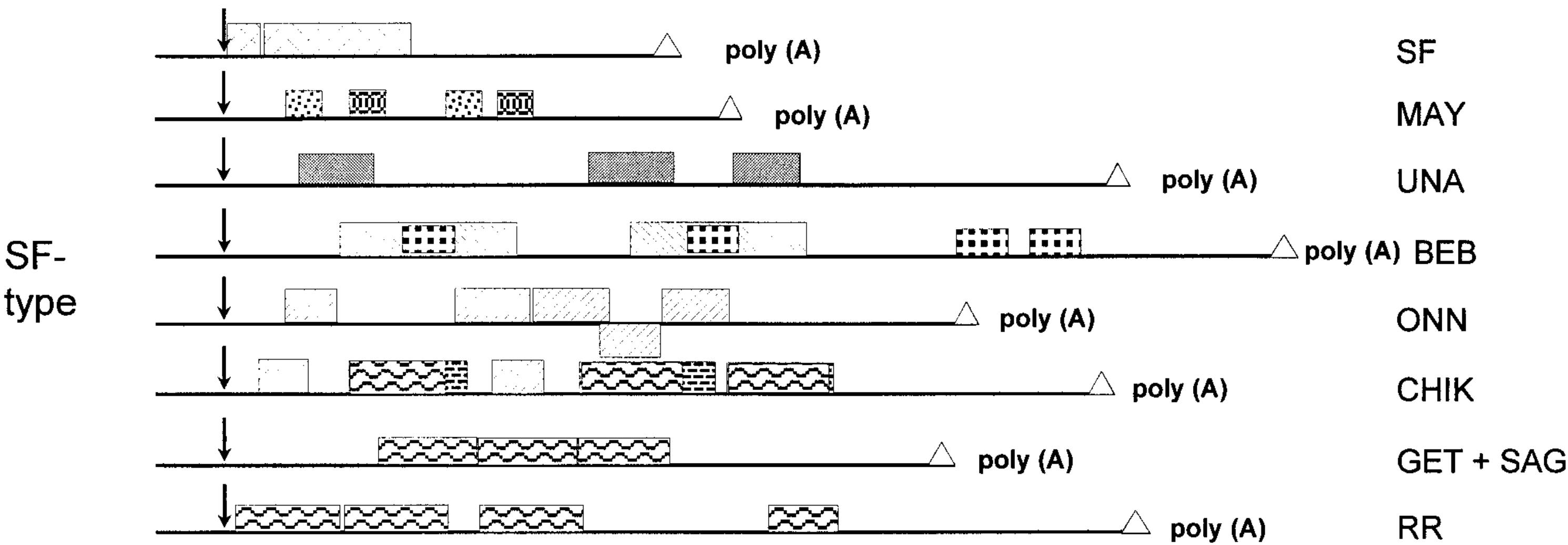
A-U	G-C	G-U
U-A	C-G	U-G

CHIKV 3'UTR: Sequence Repeats

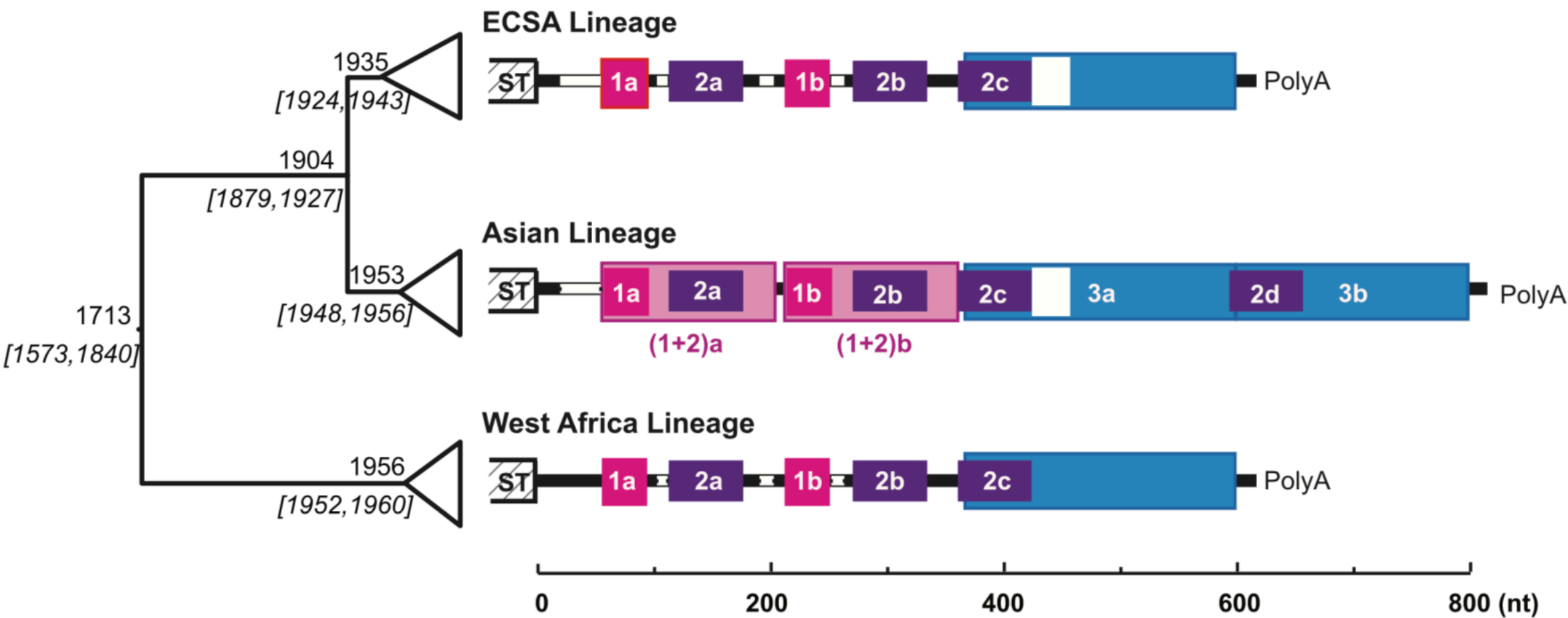


Pfeffer et al., Virology (1998)
Repeat Sequence Elements

CHIKV 3'UTR: Sequence Repeats

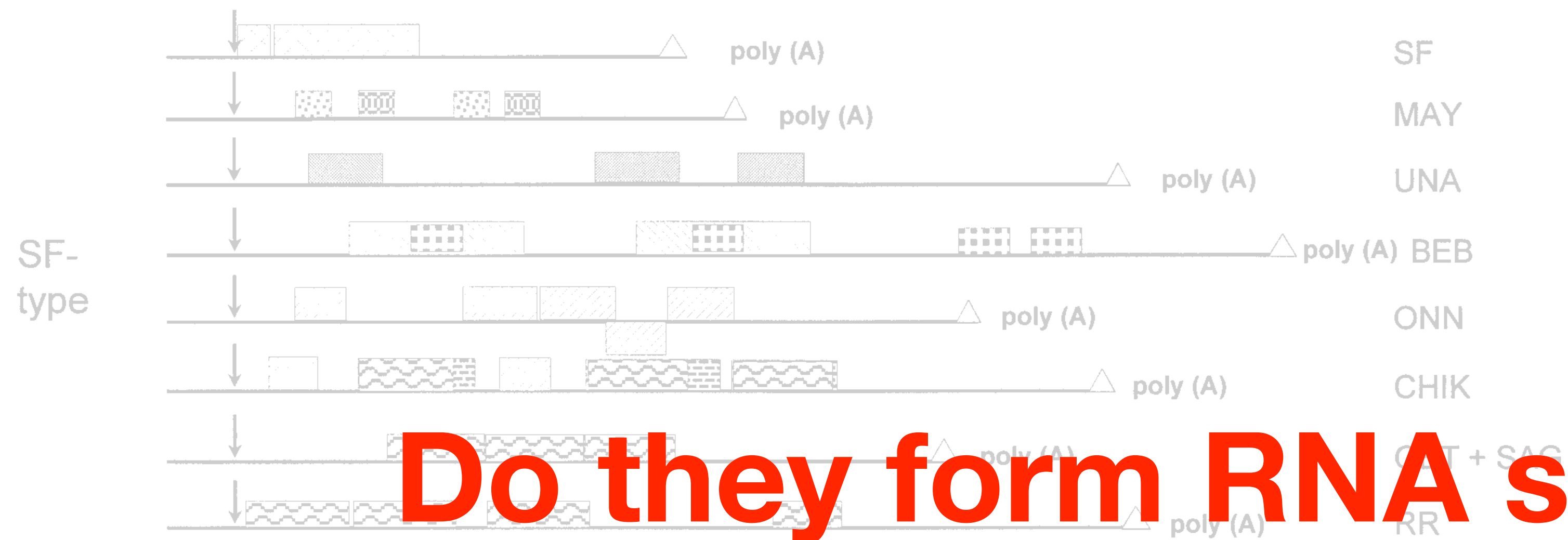


Pfeffer et al., Virology (1998)
Repeat Sequence Elements (RSEs)



Chen et al., PLoS Pathog. (2013)
Lineage-specific Direct Repeats (DRs)

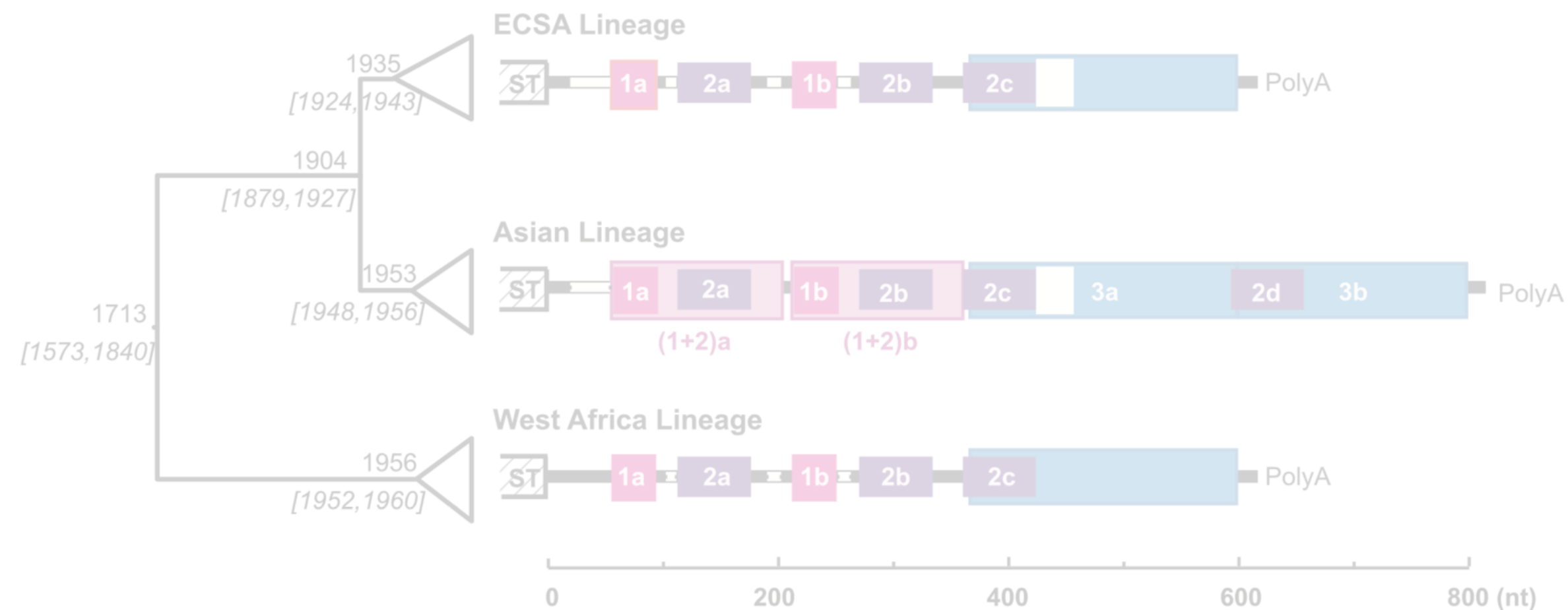
CHIKV 3'UTR: Sequence Repeats



Pfeffer et al., Virology (1998)

Repeat Sequence Elements (RSEs)

Do they form RNA structures ?



Chen et al., PLoS Pathog.(2013)

Lineage-specific Direct Repeats (DRs)

CHIKV 3'UTR



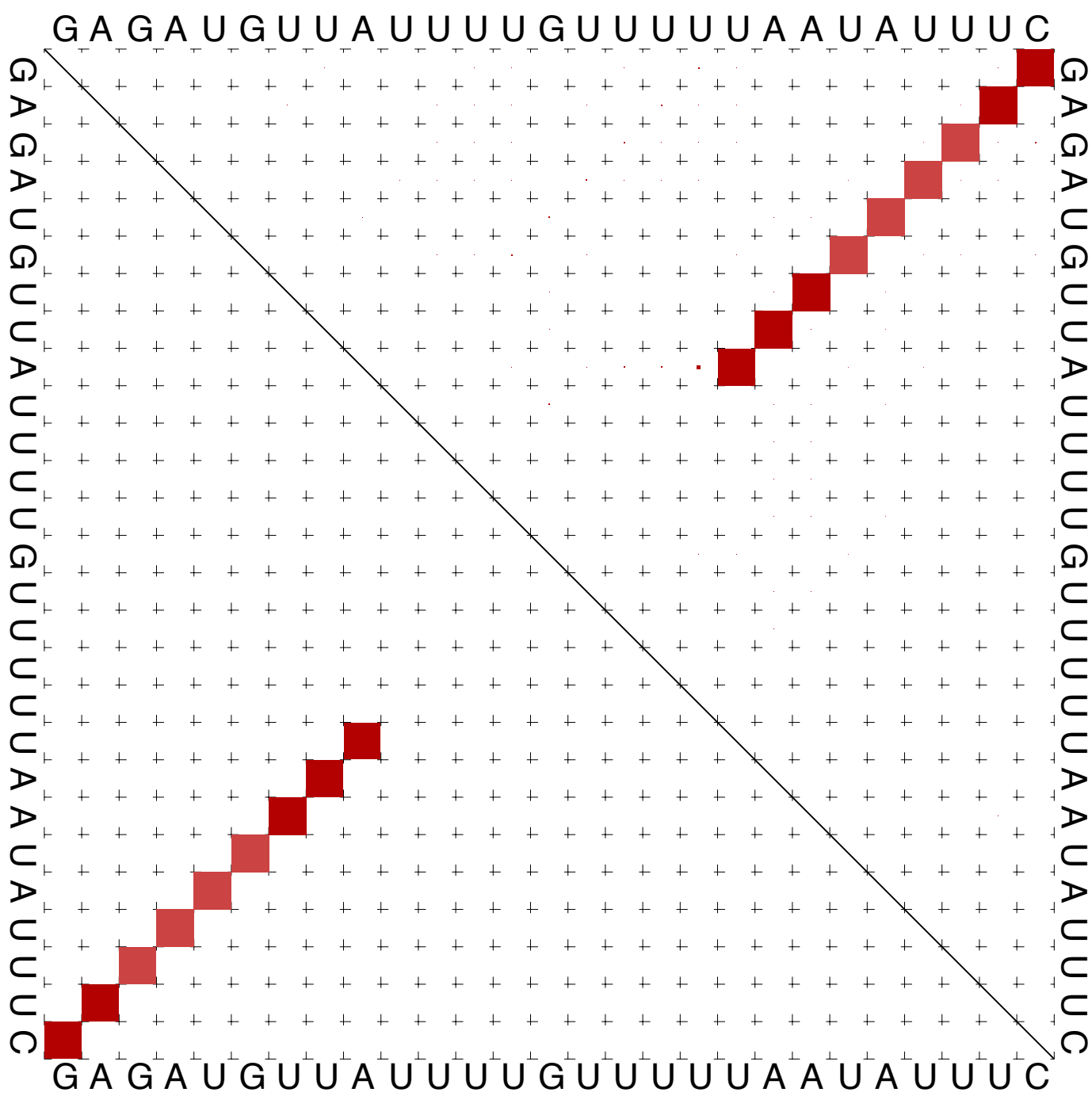
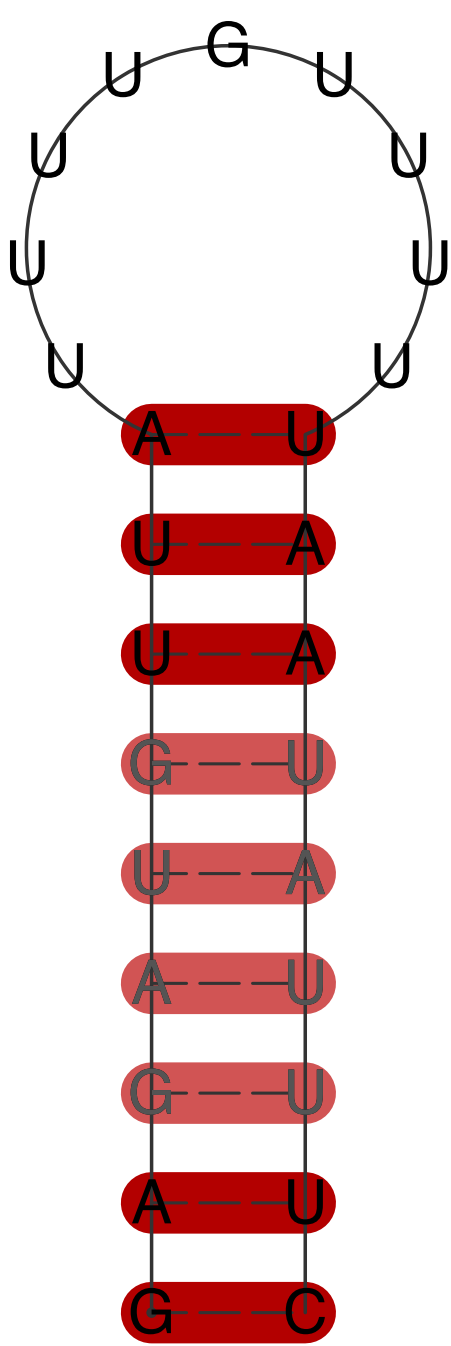
- 110 full length CHIKV 3'UTRs
- Variable length (510nt — 930nt)
- Structural alignments + Covariance models
- Thermodynamic modelling based on ViennaRNA Package

CHIKV 3'UTR: Conserved RNA Structures

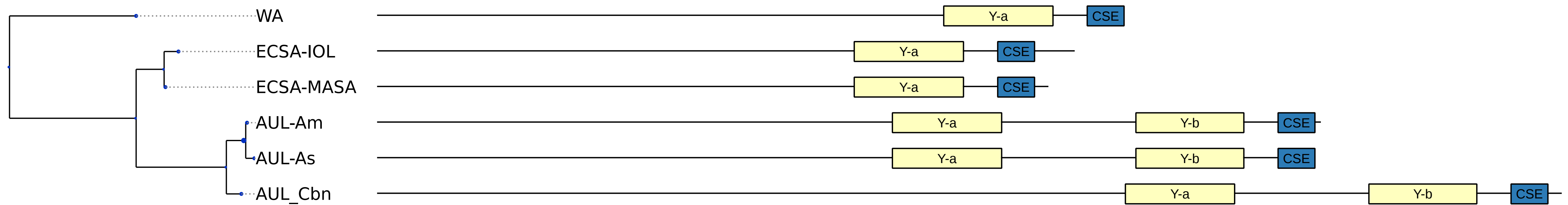


Conserved Sequence Element (CSE)

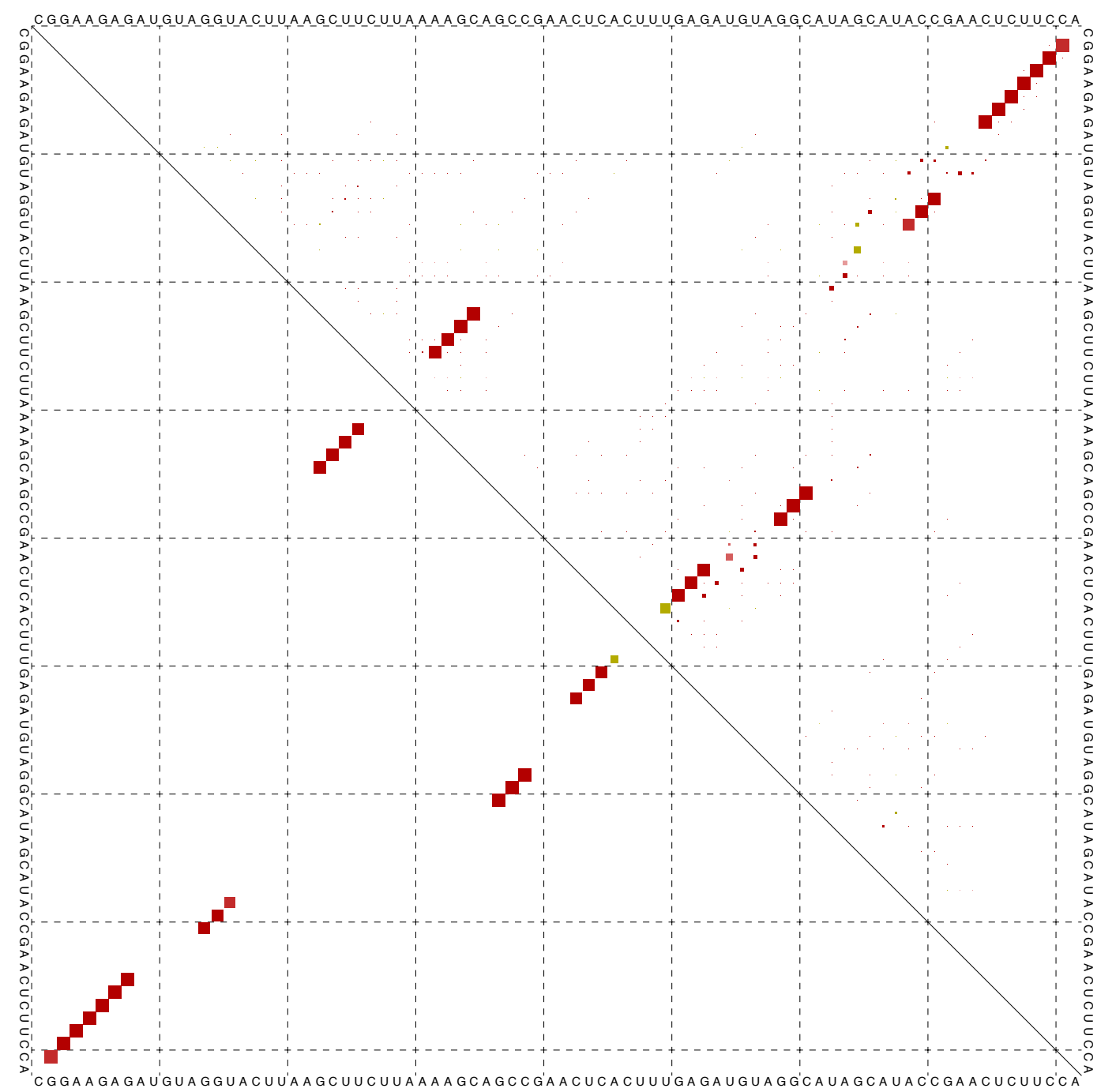
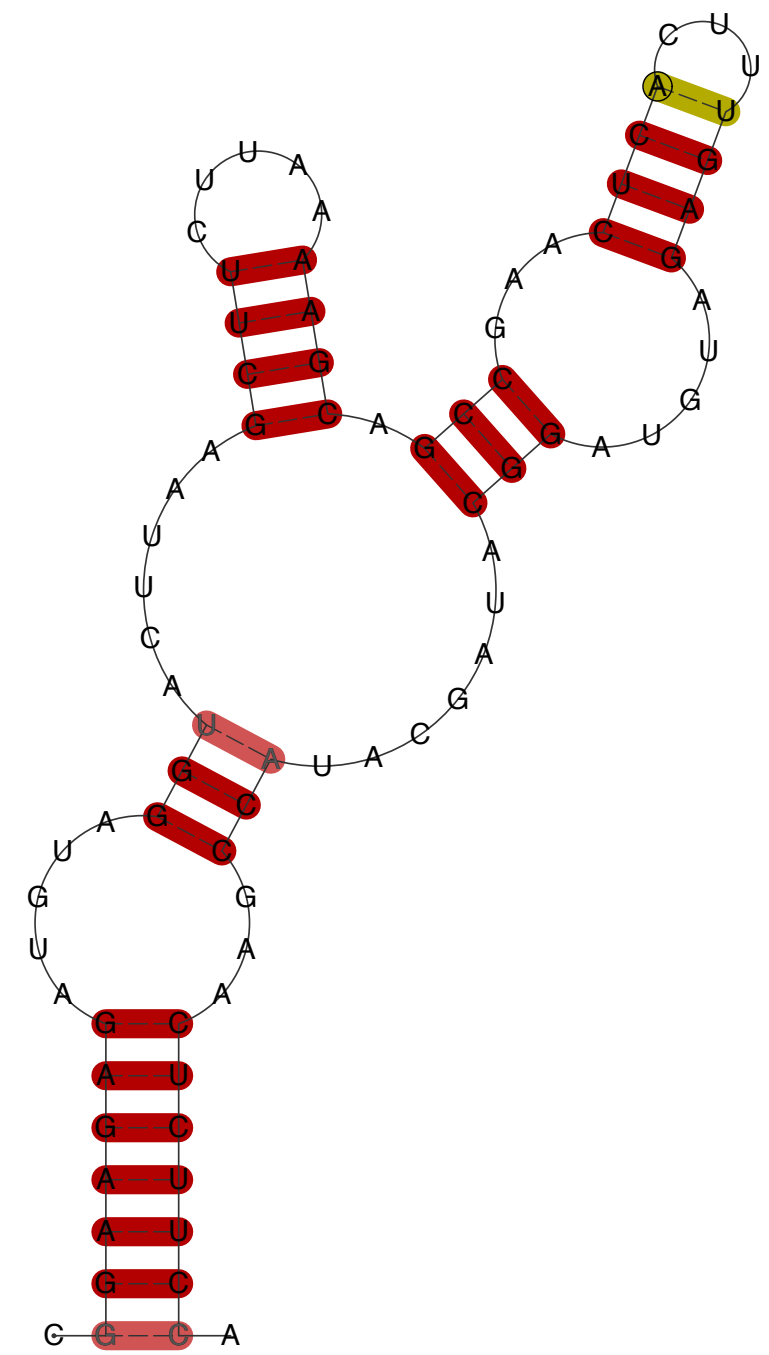
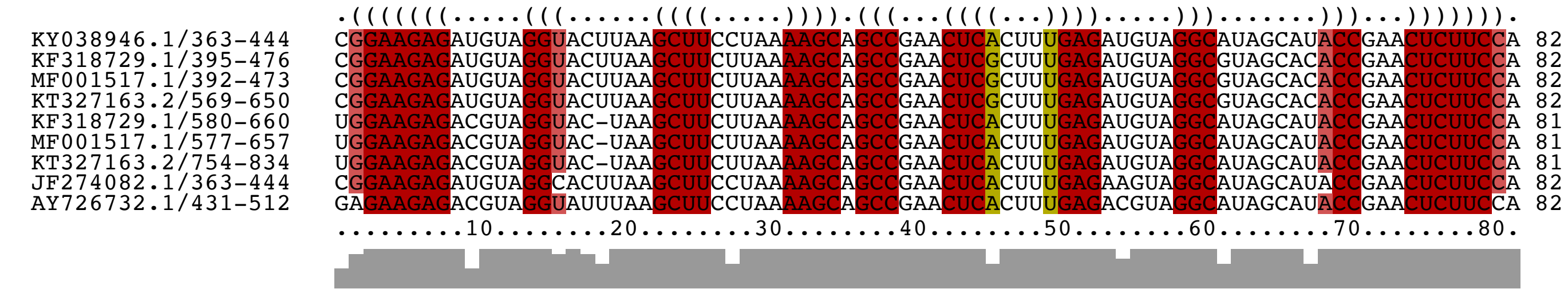
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JF274082.1/472-498	GAGAUGUUAUUUUGUUUUUAAUAUUUC	27
KY038946.1/472-498	GAGAUGUUAUUUUGUUUUUAAUAUUUC	27
AY726732.1/540-566	GAGAUGUUAUUUUGUUUUUAAUAUUUC	27
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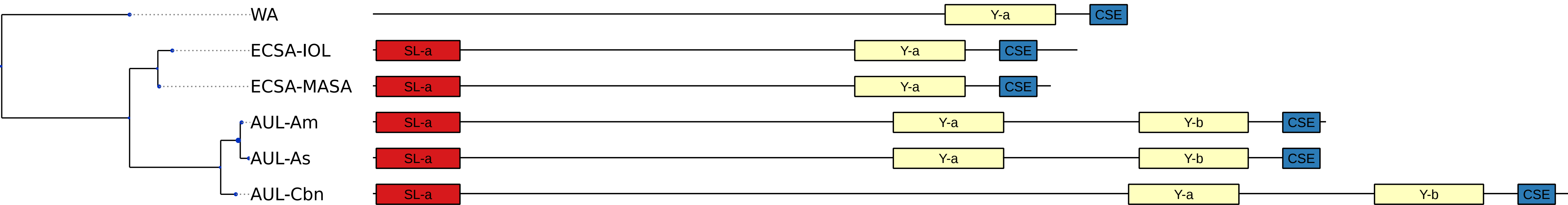
CHIKV 3'UTR: Conserved RNA Structures



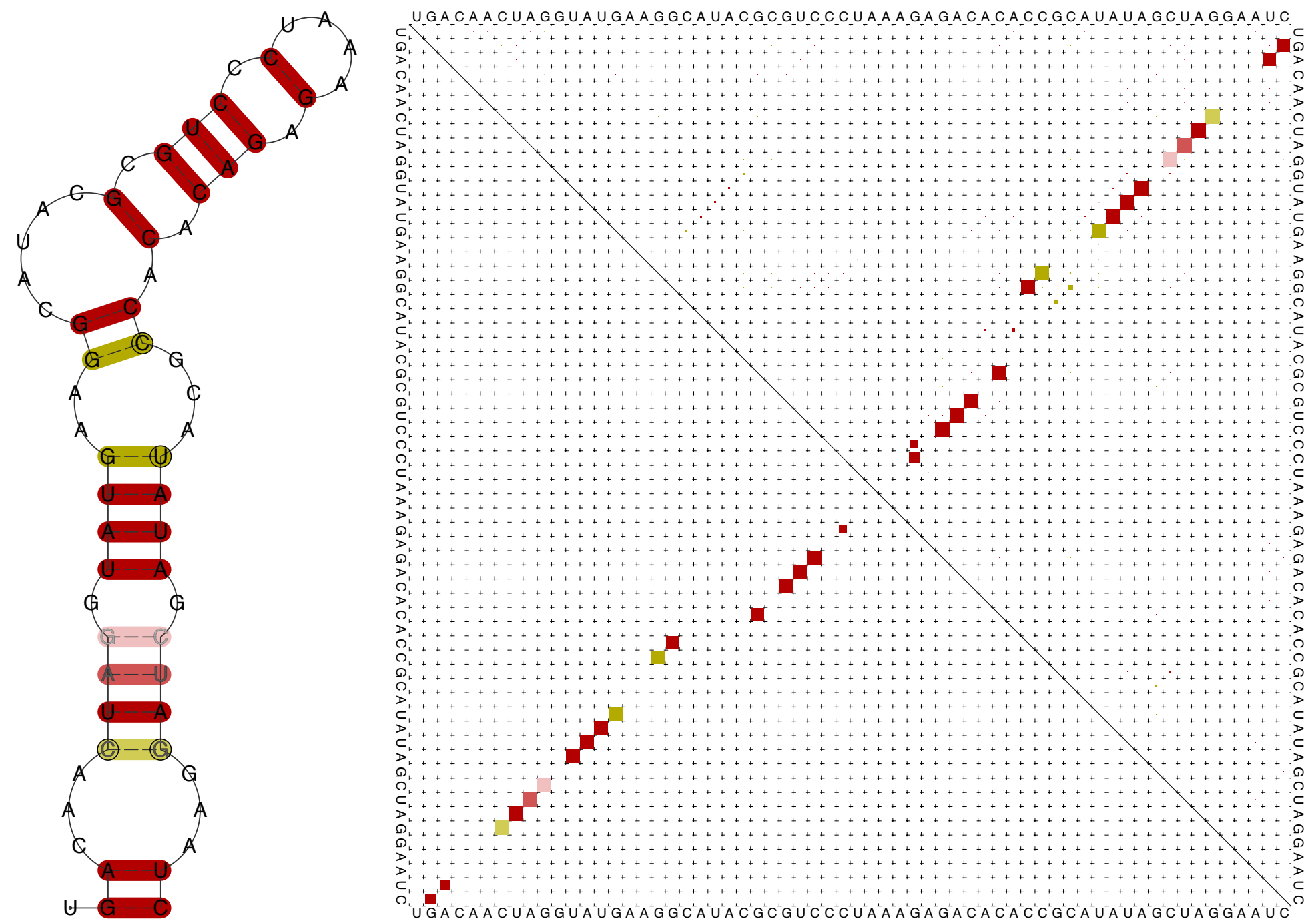
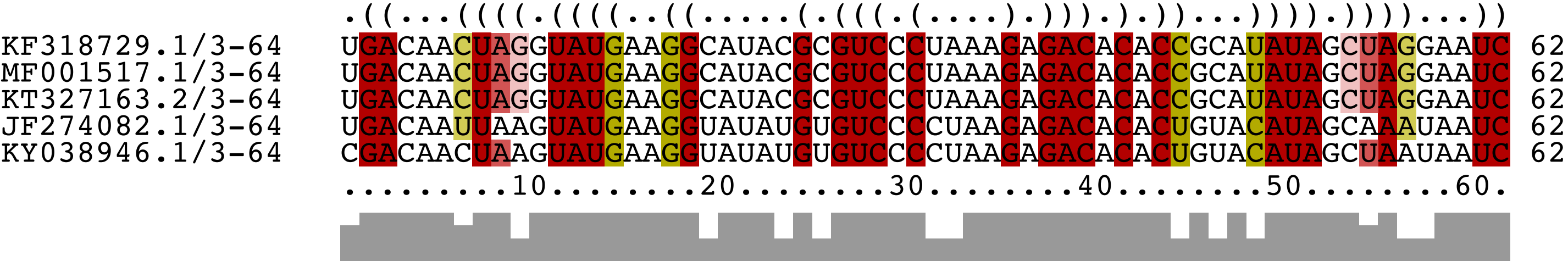
Y-shaped Element (SL-Y)



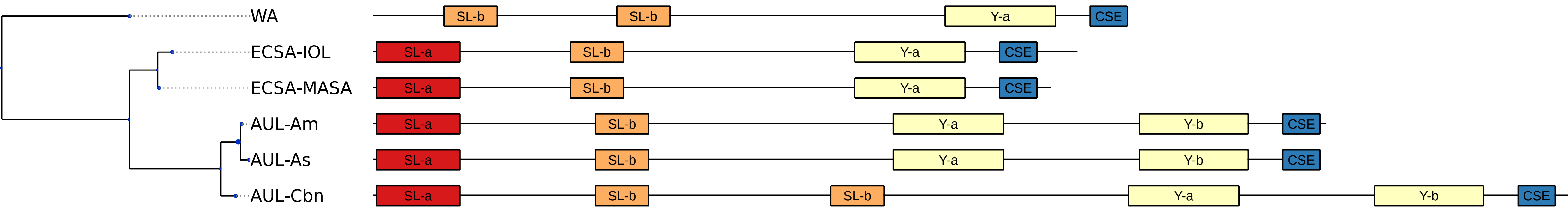
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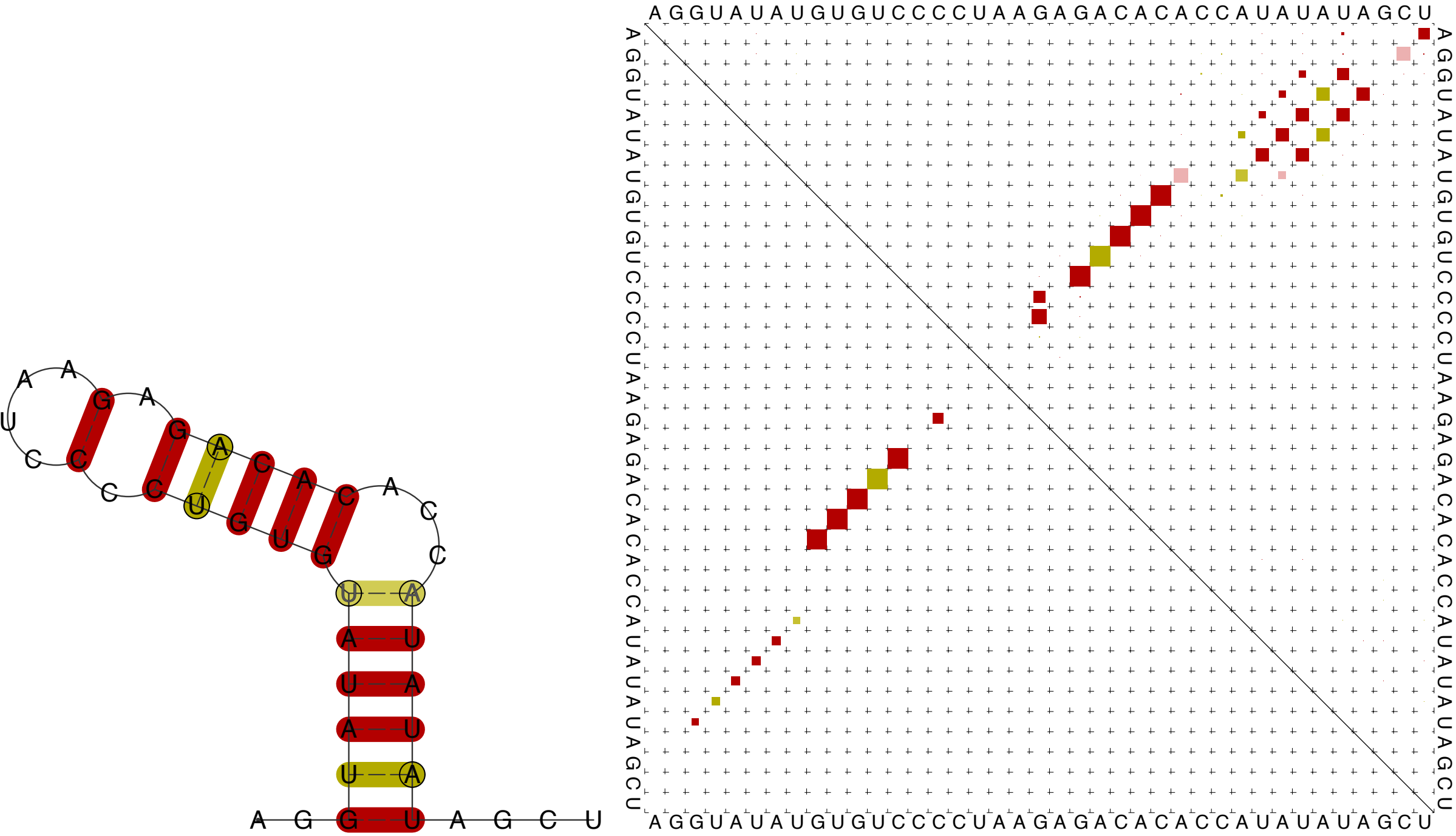
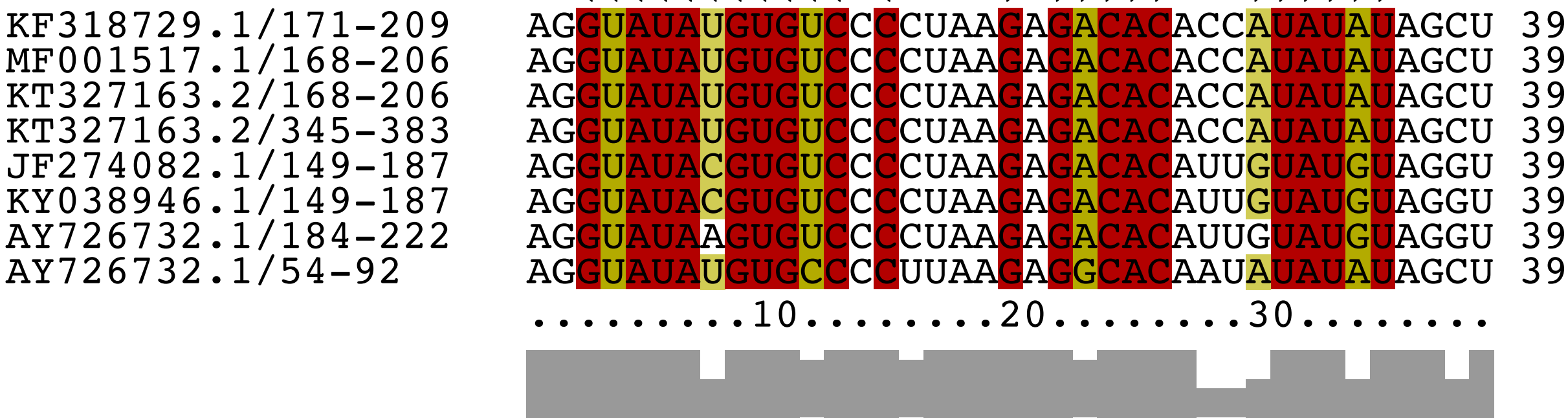
Stem-loop a (SL-a)



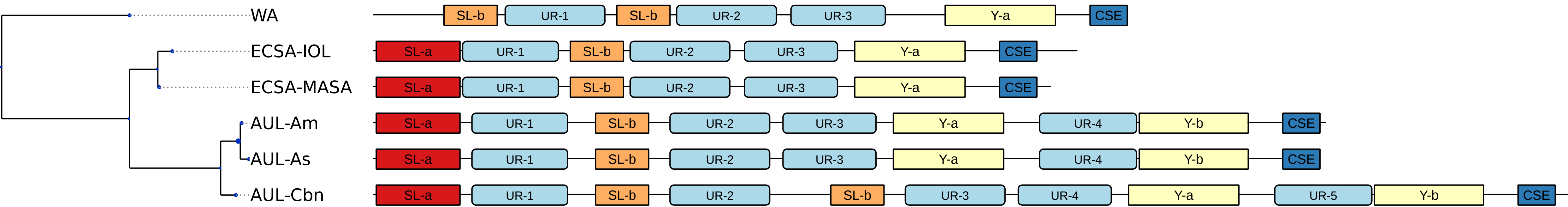
CHIKV 3'UTR: Conserved RNA Structures



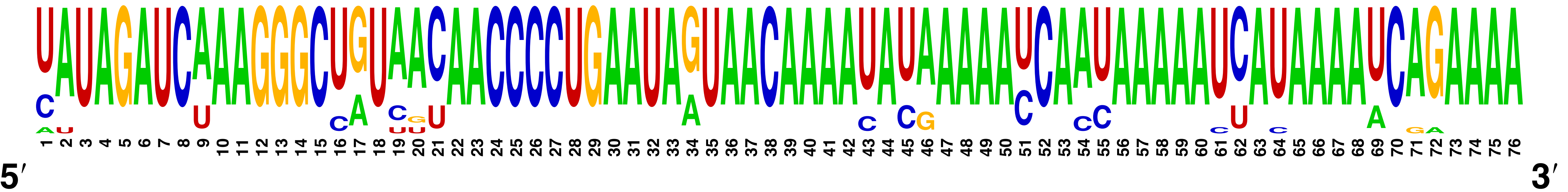
Stem-loop b (SL-b)



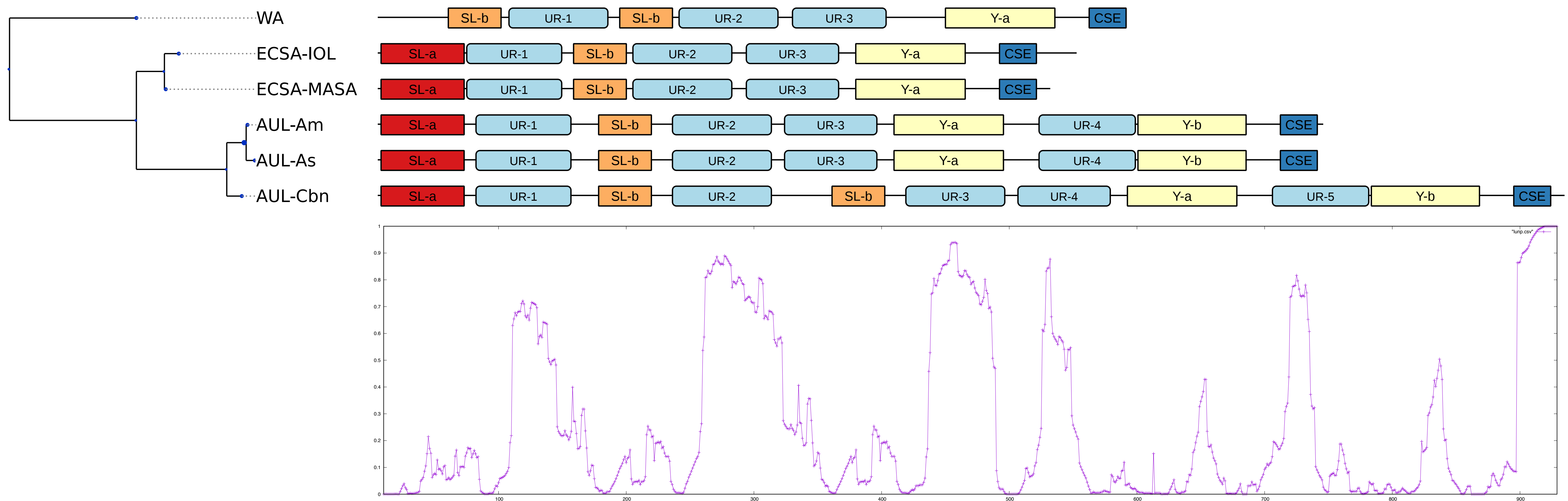
CHIKV 3'UTR: Conserved RNA Elements



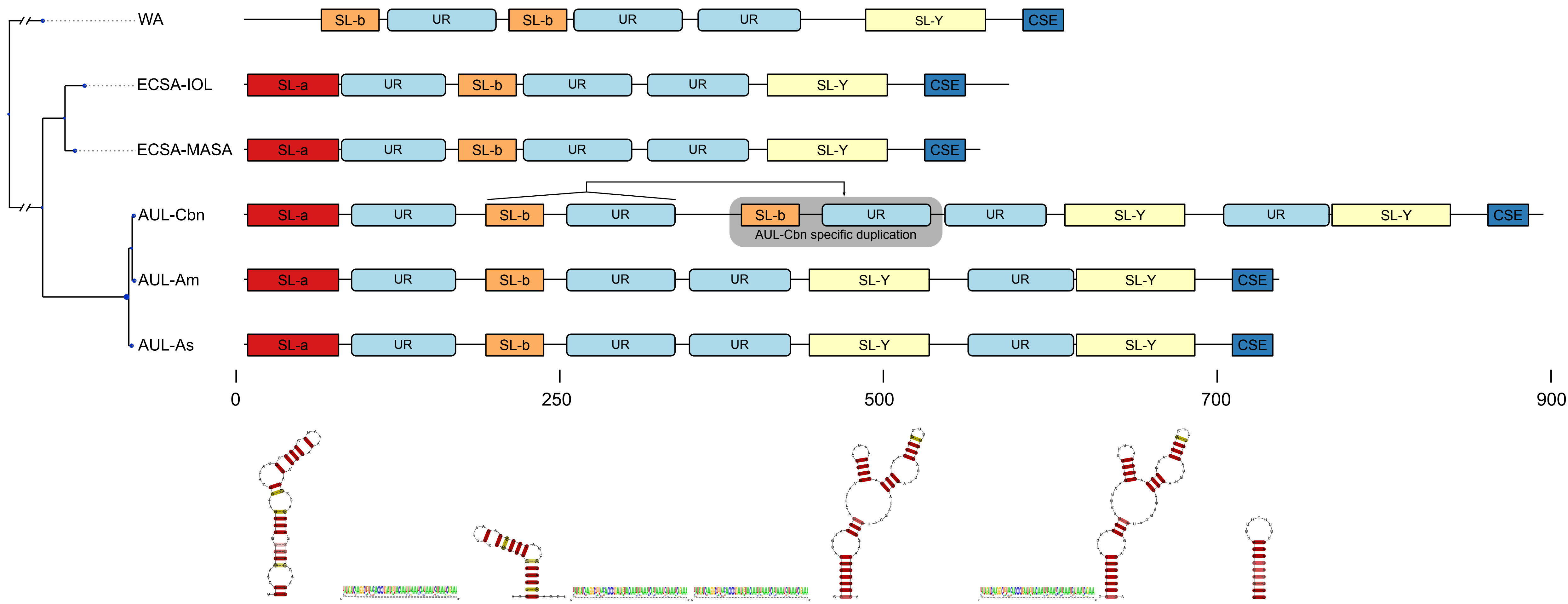
Unstructured repeat (UR)



CHIKV-Cbn Accessibility



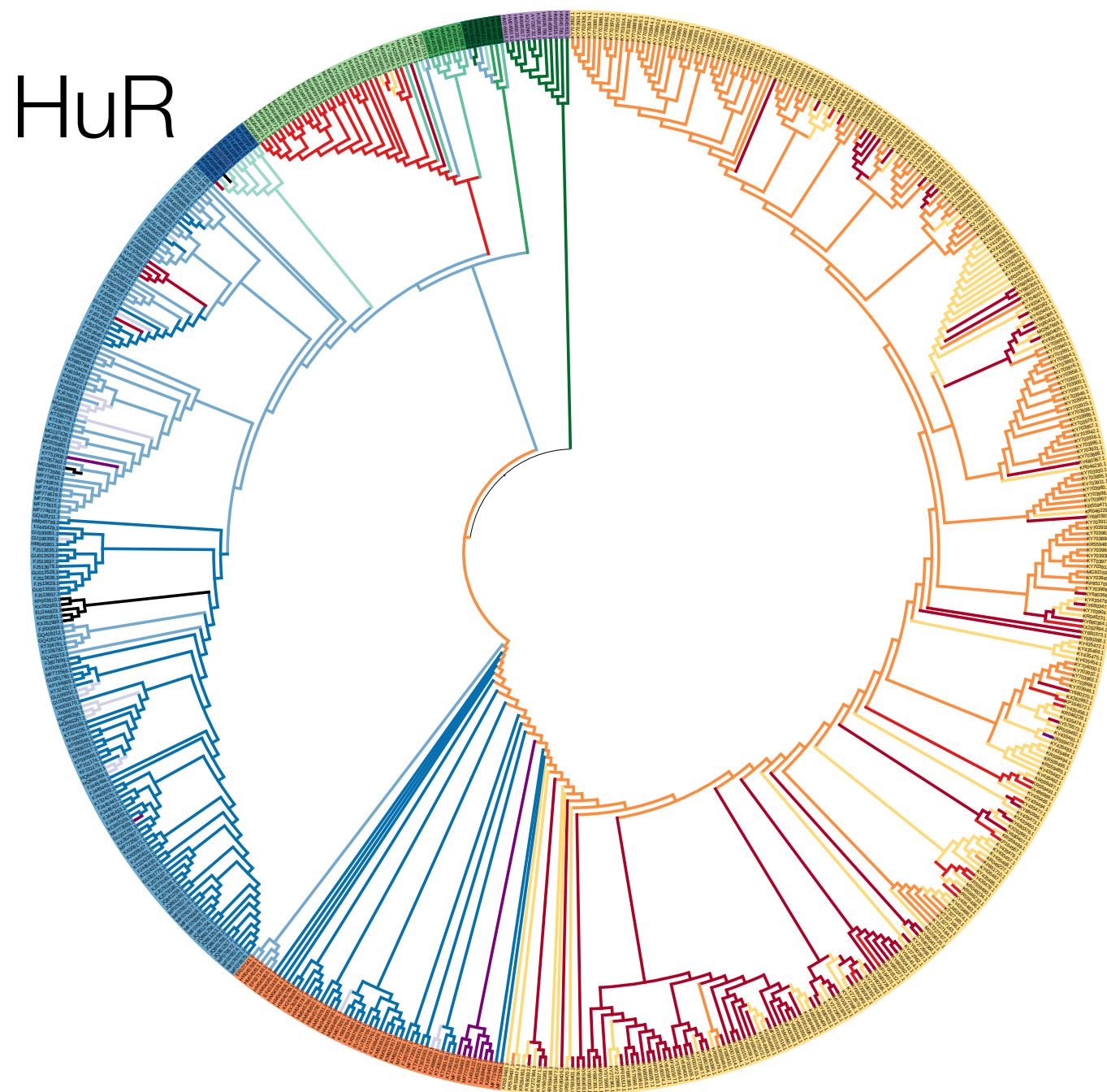
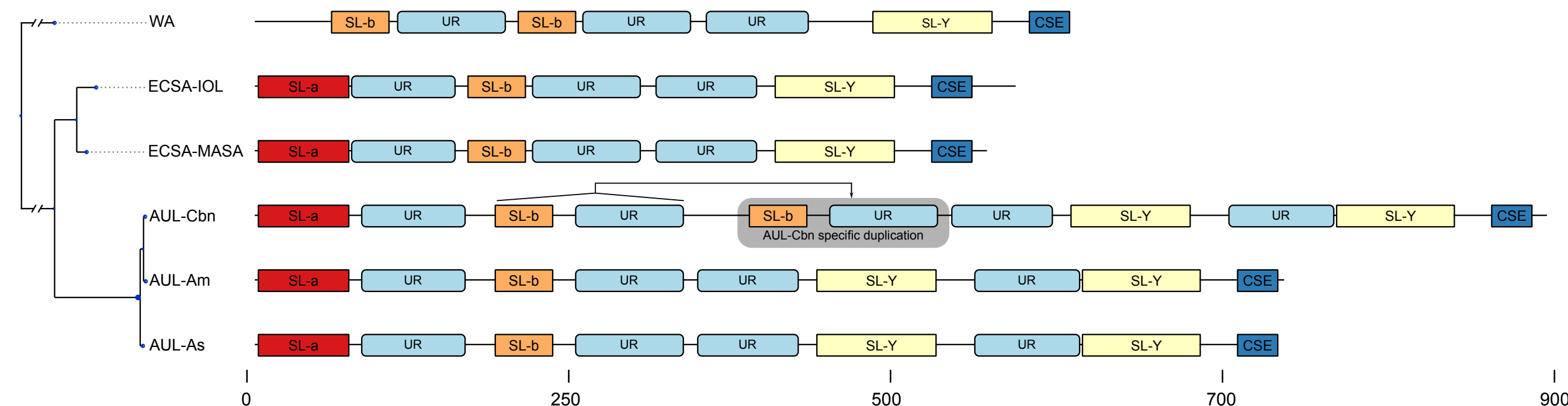
CHIKV 3'UTR Overview



Summary

- Lineage-specific non-coding RNA architecture
- Conclusive mapping of RSEs / DRs to structured and unstructured regions
- Individual lineages show varied replication potential in mosquito / vertebrate cells
- CHIKV 3'UTR stem-loops do not stall exoribonucleases
- A-rich unstructured repeats are likely sponge RNA-binding proteins like HuR
- Whether or not miRNAs are encoded needs to be established

CHIKV 3'UTRs



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Updated phylogeny of Chikungunya virus reveals lineage-specific non-coding RNAs

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